

Multimodal Fusion o A Comprehe

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低质量数据上的多模 查

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Abstract—Multimodal fusion focuses on integrating information from has achieved remarkable progress in a wide range of scenarios, inc reliability of multimodal fusion remains largely unexplored especially challenges and recent advances of multimodal fusion in the wild and view, we identify four main challenges that are faced by multimodal fu contaminated with heterogeneous noises, (2) incomplete multimodal data that the qualities or properties of different modalities are significa of each modality dynamically changes with respect to different sampl state of the field and identify several potential directions. We also pr interesting future research directions.

Index Terms—Multimodal fusion, machine learning, robustness.

摘要—多模态融合专注于整合来自多种模态的信息, 以实现更准确的而, 多模态融合的可靠性在很大程度上仍未得到充分探索, 尤其是在战和最新进展, 并对其进行全面的分类。从数据为中心的观点来看, (1) 被异构噪声污染的噪声多模态数据, (2) 某些模态缺失的不完态数据, 以及 (4) 各模态质量随不同样本动态变化的质量变化多模态潜在方向。我们还对该领域的开放问题进行了讨论, 并提出了有趣的

索引词—多模态融合, 机器学习, 鲁棒性。

1 INTRODUCTION

OUR perception of the world is based on multiple modalities, e.g., touch, sight, hearing, smell and taste. Even when some sensory signals are unreliable, humans can extract useful clues from imperfect multimodal inputs and further piece together the whole scene of happening events [1]. With the development of sensory technology, we can easily collect diverse forms of data for analysis. To fully release the value of each modality, multimodal fusion emerges as a promising paradigm to obtain precise and reliable predictions by integrating all available cues for downstream analysis tasks, e.g., medical image analysis, autonomous vehicles [2], [3] and sentiment recognition [4], [5], [6]. Intuitively, fusing information from different modalities offers the possibility of exploring cross-modal correlation and gaining better performance. However, there is growing recognition that widely-used AI models are often misled by spurious correlations and biases within low-quality data. In real-world setting, the quality of different modalities usually varies due to unexpected environmental factors or sensor issues. Some recent studies have shown both empirically and

1 引言

对世界的感知基于多种模态, 例如触觉、视觉、听觉、嗅觉和味觉。即使某些感觉信号不可靠, 人类也能从不完美的多模态输入中提取有用线索, 并进一步拼凑出发生事件的整体场景 [1]。随着感觉技术的发展, 我们可以轻松收集各种形式的数据进行分析。为了充分释放每种模态的价值, 多模态融合已成为一个有前景的范式, 通过整合所有可用线索为下游分析任务获得精确和可靠的预测, 例如医学图像分析、自动驾驶 [2], [3] 和情感识别 [4], [5][6]。直观地讲, 融合来自不同模态的信息提供了探索跨模态相关性的可能性, 并获得更好的性能。然而, 人们越来越认识到, 广泛使用的AI模型常常被低质量数据中的虚假相关性和偏差所误导。在现实场景中, 由于意外的环境因素或传感器问题, 不同模态的质量通常会有所不同。最近的一些研究表明, 这在经验和

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on Low-quality Data: nsive Survey

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m multiple modalities with the goal of more accurate prediction, which
, including autonomous driving and medical diagnosis. However, the
ially under low-quality data settings. This paper surveys the common
nd presents them in a comprehensive taxonomy. From a data-centric
l fusion on low-quality data, namely (1) noisy multimodal data that are
dal data that some modalities are missing, (3) imbalanced multimodal
ficantly different and (4) quality-varying multimodal data that the quality
mples. This new taxonomy will enable researchers to understand the
o provide discussion for the open problems in this field together with



theoretically that conventional multimodal fusion may fail on
low-quality multimodal data in the wild, e.g., imbalanced [7],
[8], [9], [10], noisy [11] or even corrupted [12] multimodal
data. To overcome such a limitation and take a step to strong
and generalized multimodal learning in the real world, we
identify the properties of low-quality multimodal data and
focus on some unique challenges for multimodal machine
fusion in real-world setting. We also highlight technical
advances that may help to make multimodal fusion more
reliable and trustworthy in the open environment. In this
paper, we identify and explore four core technical challenges
surrounding multimodal fusion on low-quality multimodal
data. They are summarized as follows:

- (1) **Noisy multimodal data.** The first fundamental challenge is learning how to mitigate the underlying influence of arbitrary noise in multimodal data. High-dimensional multimodal data tend to contain complex noise. The heterogeneity of multimodal data makes it challenging while also provides chances to identify and reduce the potential noise by exploring the correlation among different modalities.
- (2) **Incomplete multimodal data.** The second fundamental challenge is learning with incomplete multimodal data. For instance, in the medical field, patients even with the same disease may choose different medical examinations producing incomplete multimodal data. Developing flexible and reliable multimodal learning methods that can handle incomplete multimodal data is a challenging yet promising research direction.
- (3) **Imbalanced multimodal data.** The third fundamental challenge is how to mitigate the influence of bias and discrepancy between modalities. For example, the visual modality is more effective than the audio modality on the whole, leading the model to take shortcuts and lack exploration of audio. Although existing fusion

态融合：一项综合调

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的预测，这在自动驾驶和医学诊断等广泛场景中已取得显著进展。然
在低质量数据设置下。本文调查了多模态融合在现实世界中的常见挑
，我们识别出在低质量数据上多模态融合面临的四个主要挑战，即
完整多模态数据，（3）不同模态的质或属性显著不同的不平衡多模
态数据。这一新分类将帮助研究人员了解该领域的现状并确定几个
的未来研究方向。



理论上，传统的多模态融合在现实世界中的低质量多模态数
据上可能会失败，例如不平衡的 [7]、噪声的 [8], [9], [10], 甚
至损坏的 [11] 多模态数据。为了克服这种局限性，并迈向现
实世界中的强健和泛化的多模态学习，我们识别了低质量多
模态数据的特性，并关注现实世界设置中多模态机器融合的一
些独特挑战。我们还强调了可能有助于使多模态融合在开
放环境中更加可靠和值得信赖的技术进步。在本文中，我们
识别并探讨了围绕低质量多模态数据上的多模态融合的几个
核心技术挑战。它们总结如下：

- (1) **噪声多模态数据。** 第一个基本挑战是学习如何减轻多模
态数据中任意噪声的潜在影响。高维多模态数据往往包含复
杂的噪声。多模态数据的异质性使其具有挑战性，同时也提
供了机会，通过探索不同模态之间的相关性来识别和减少潜
在的噪声。
- (2) **不完全多模态数据。** 第二个基本挑战是在不完全多模态
数据中进行学习。例如，在医疗领域，即使患有相同疾病的
患者也可能选择不同的医学检查，从而产生不完全多模态数
据。开发能够处理不完全多模态数据的灵活且可靠的多模态
学习方法是一个具有挑战性但充满前景的研究方向。
- (3) **不平衡的多模态数据。** 第三个基本挑战是如何减轻模态
之间的偏差和不一致的影响。例如，在整体上，视觉模态比
音频模态更有效，导致模型走捷径并缺乏对音频的探索。尽
管现有的融合

methods exhibit promising performance, they may fail to perform better than the unimodal predominant model for inference on some modality-preferred applications.

- (4) **Quality dynamically varying multimodal data.** The fourth fundamental challenge is how to adapt the quality dynamically varying nature of multimodal data. In practice, the quality of one modality often varies for different samples due to unforeseeable environmental factors or sensor issues. For example, the RGB image is less informative than the thermal modality in low-light or backlight conditions. Therefore, dynamically integrating multimodal data by being aware of the varying quality is necessary in real-world applications.

Towards addressing these increasingly important issues of multimodal fusion, this study systematically organizes the key challenges through several taxonomies. Different from previous related works discuss about various multimodal learning tasks [13], [14], this survey is mainly focusing on multimodal fusion which is the most fundamental problem in multimodal learning and the unique challenges raising upon low-quality multimodal data in downstream tasks, including clustering, classification, object detection and semantic segmentation. In the following sections, we detailed this field by recent advances and technical challenges of multimodal fusion facing low-quality data: learning on noisy multimodal data (Section 2), missing modality imputation (Section 3), balanced multimodal fusion (Section 4), and dynamic multimodal fusion (Section 5). A discussion in Section 6 is provided as a conclusion.

2 LEARNING ON NOISY MULTIMODAL DATA

Collecting high-quality multimodal data in real-world scenarios inevitably faces significant challenges due to the presence of noise. Multimodal data [15] noise can arise from sensor errors [16], environmental interference, or transmission losses. For vision modality, the electronic noise in sensors leads to detail loss. Besides, audio modality may suffer from unexpected distortion due to environmental factors. To make things worse, weakly aligned or even unaligned multimodal samples are also commonly presented, which can be seen as more complex noise that lies in a higher-level semantic space. Fortunately, considering the correlation between multiple modalities or better utilizing multimodal data can help the fusion of noisy multimodal data. Various related works [16], [17], [18] demonstrate that multimodal models surpass their unimodal counterparts. This can be attributed to the ability of multimodal data to identify and mitigate potential noise by leveraging the correlations among different modalities.

Multimodal noise can be roughly split into two categories according to their source: 1) modal-specific noise that arises from sensor errors, environmental factors, or transmission of each modality separately, 2) cross-modal noise that arises from weakly aligned or unaligned multimodal pairs which can be regarded as semantic level noise.

2.1 Modal-specific noise reduction

Modal-specific noise reduction methods highly depends on the input modalities and task at hand. In this section, we focus on **visual** noise reduction in multimodal image

these methods表现出良好的性能，但在某些偏好特定模态的应用中进行推理时，它们可能无法比单模态主导模型表现得更好。

- (4) **动态变化质量的多元模态数据。**第四个基本挑战是如何适应多元模态数据的动态变化质量。在实践中，由于不可预见的环境因素或传感器问题，一种模态的质量往往因不同样本而异。例如，在低光或逆光条件下，RGB图像的信息量不如热成像模态丰富。因此，在现实世界应用中，需要根据质量变化动态整合多元模态数据。

为应对这些日益重要的多元模态融合问题，本研究通过多个分类体系系统地组织了关键挑战。与之前讨论各种多元模态学习任务的相关工作不同，本综述主要聚焦于多元模态融合——这是多元模态学习中最基本的问题，以及下游任务中低质量多元模态数据所面临的独特挑战，包括聚类、分类、目标检测和语义分割。在后续章节中，我们将通过近期进展和技术挑战详细阐述多元模态融合领域面对低质量数据的情况：含噪声多元模态数据学习（第2节）、缺失模态插补（第3节）、均衡多元模态融合（第4节）和动态多元模态融合（第5节）。第6节提供了讨论作为结论。

2 在噪声多模态数据上的学习

在现实场景中收集高质量的多模态数据不可避免地会面临重大挑战，这是由于噪声的存在。多模态数据 [15] 噪声可能源于传感器错误 [16]、环境干扰或传输损耗。对于视觉模态，传感器中的电子噪声会导致细节丢失。此外，音频模态也可能由于环境因素而遭受意外的失真。更糟糕的是，弱对齐甚至不对齐的多模态样本也很常见，这可以被视为存在于更高语义空间中的更复杂的噪声。幸运的是，考虑多个模态之间的相关性或更好地利用多模态数据可以帮助融合噪声多模态数据。各种相关工作 [16][17], [18] 表明，多模态模型优于它们的单模态对应物。这归因于多模态数据能够通过利用不同模态之间的相关性来识别和减轻潜在的噪声。

根据其来源，多模态噪声大致可分为两类：1) 模态特定噪声，它源于传感器错误、环境因素或每个模态的单独传输，2) 跨模态噪声，它源于弱对齐或不对齐的多模态对，这些可以被视为语义级噪声。

2.1 模态特定噪声抑制

模态特定的噪声降低方法高度依赖于输入模态和具体任务。在本节中，我们专注于视觉噪声降低在多模态图像融合任务中，展示一个案例。大多数模态特定的去噪方法专注于从多模态数据中聚合有用信息，并减轻噪声在多模态融合中的影响。

fusion task to present a showcase. Most modal-specific denoise methods focus on aggregating useful information from multimodal data and mitigating the influence of noise in multimodal fusion.

2.1.1 Joint variation based fusion

Joint optimization is often used in variation-based noise reduction when fusing multiple visual modalities, e.g., RGB and thermal. In unimodal noise reduction field, taking image modality as an example, classical denoise model based on Total Variation (TV) is equivalent to solving the following optimization problems:

$$\min \iint [u(x, y) - u_0(x, y)]^2 dx dy + \lambda \iint |\nabla u(x, y)| dx dy, \quad (1)$$

where $u_0(x, y)$, $u(x, y)$ represent the input pixel and output pixel after noise reduction respectively. For multimodal noise reduction, the joint variational model for fusion and denoise is defined as follows:

$$\min \iint \sum_{m=1}^M w_m [u(\mathbf{x}) - u_m(\mathbf{x})]^2 d\mathbf{x} + \lambda \iint |\nabla u(\mathbf{x})| d\mathbf{x}, \quad (2)$$

where $u_m(\mathbf{x})$ represents the input data from modality m , w_m is the non-negative weight satisfying $\sum_{m=1}^M w_m = 1$. In a nutshell, joint optimization is the solution of a specific optimization problem using data from different modalities. Wang *et al.* [19] proposed a variational model in the pixel domain and wavelet domain for joint fusion and denoise of noisy multifocus images. Kumar *et al.* [20] also employed the total variational model to fuse images acquired using multiple sensors and achieves excellent results on two modal images provided in the medical field and aircraft navigation field. Recently, Padmavathi *et al.* [21] proposed an image fusion algorithm for constructing a fused image using the total variation model. They incorporated an optimized adaptive weighting fusion scheme for medical multimodal data analysis (i.e., Magnetic Resonance images and Positron Emission Tomography images). Nie *et al.* [22] introduced a total variation-based fusion method for infrared and visible images. The method leverages the total variation model to fuse two modalities, enabling enhanced image quality and information integration. Quan *et al.* [23] developed a feature fusion method called relative total variation structure analysis (RTVSA) for urban area classification. This approach combines various features derived from hyperspectral imaging (HSI) and LiDAR data, providing a comprehensive analysis for accurate classification. Furthermore, Liu *et al.* [24] designed a two-stage enhancement (TSE) framework for infrared and visible image fusion. Their framework incorporated an attention mechanism and a feature-linking model (FLM), combining structure adaptive total-variational (SATV) and L1 sparsity terms to extract two-scale detail layers and a base layer. The proposed method demonstrates robustness and effectiveness in improving the quality of fused images.

2.2 Cross-modal noise reduction

Many multimodal tasks (e.g., multimodal object detection, vision-and-language understanding) highly depends on

融合任务以展示一个案例。大多数模态特定的去噪方法专注于从多模态数据中聚合有用信息，并减轻噪声在多模态融合中的影响。

2.1.1 基于联合变化融合

联合优化常用于基于变化的噪声抑制中，当融合多个视觉模态（例如RGB和热成像）时。在单模态噪声抑制领域，以图像模态为例，基于总变化（TV）的经典去噪模型相当于解决以下优化问题：

$$\min \iint [u(x, y) - u_0(x, y)]^2 dx dy + \lambda \iint |\nabla u(x, y)| dx dy, \quad (1)$$

其中 $u_0(x, y)$ $u(x, y)$ 分别表示噪声抑制前后的输入像素和输出像素。对于多模态噪声抑制，融合和去噪的联合变化模型定义如下：

$$\min \iint \sum_{m=1}^M w_m [u(\mathbf{x}) - u_m(\mathbf{x})]^2 d\mathbf{x} + \lambda \iint |\nabla u(\mathbf{x})| d\mathbf{x}, \quad (2)$$

其中 $u_m(\mathbf{x})$ 表示来自模态 m 的输入数据，

w_m 是满足 $\sum_{m=1}^M w_m = 1$ 的非负权重。简而言之，联合优化是使用来自不同模态的数据解决特定优化问题的方法。王等

[19] 提出了一种用于噪声多焦点图像联合融合和去噪的像素域和Wavelet域变分模型。库马尔等 [20] 也采用了总变化模型来融合使用多个传感器获取的图像，并在医学领域和航空导航领域提供的两种模态图像上取得了优异的结果。最近，Padmavathi等 [21] 提出了一种基于总变化模型的图像融合算法，用于构建融合图像。他们为医学多模态数据分析（即核磁共振图像和正电子发射断层扫描图像）引入了一种优化的自适应加权融合方案。聂等 [22] 介绍了一种基于总变化的红外和可见图像融合方法。该方法利用总变化模型融合两种模态，实现图像质量和信息集成提升。全等 [23] 开发了一种称为相对总变化结构分析（RTVSA）的城市区域分类特征融合方法。该方法结合了从高光谱成像（HSI）和激光雷达数据中提取的各种特征，为精确分类提供全面分析。此外，刘等 [24] 设计了一个用于红外和可见图像融合的两阶段增强（TSE）框架。他们的框架结合了注意力机制和特征链接模型（FLM），结合结构自适应总变化（SATV）和L1稀疏项来提取两尺度细节层和基层。所提出的方法在提高融合图像质量方面表现出鲁棒性和有效性。

2.2 跨模态噪声抑制

许多多模态任务（例如多模态目标检测、视觉与语言理解）高度依赖于

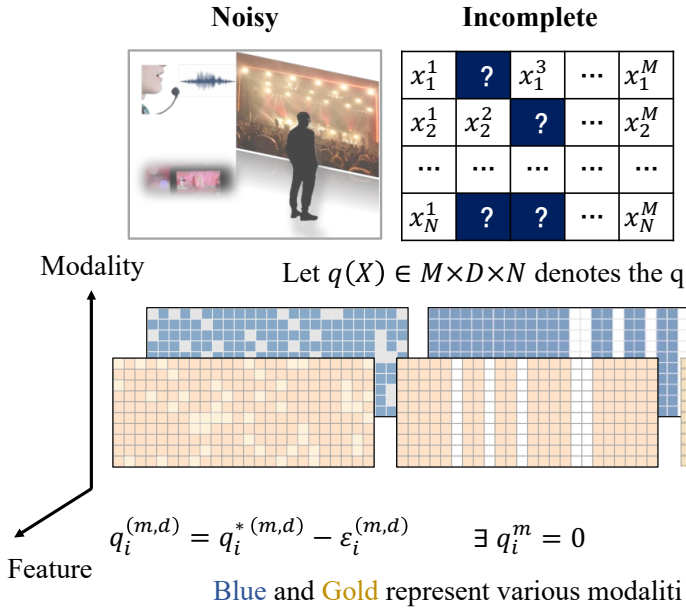


Fig. 1: Illustrations of challenges for machine learning on low modalities. Deeper color denotes higher quality. Assuming modalities, the dimension of each modality is D . $q(x)$ denote collected from x that can support the downstream tasks. (a) T by unexpected environmental factors. (b) Certain modalities convey any useful information. (c) The expected quality of m The quality of different modality are varying for samples.

correctly aligned multimodal training data. However, the real-world multimodal pairs tend to contain weakly aligned or even unaligned samples [25]. For example, in RGB-thermal multimodal object detection, multimodal inputs are often weakly aligned, i.e., the locations of the same objects may be shifted across different modalities [26]. In social media, the text descriptions are frequently irrelevant with the image content, i.e., unpaired modalities. In this section, we consider weakly aligned or unaligned multimodal samples as cross-modal noise. Compared to modal-specific noise, cross-modal noise lies in a higher-level semantic space. Current cross-modal noise reduction methods can be roughly split into rules-based filtering, model-based rectifying and noise robustness regularization.

From the data perspective, some strict rules are used to conduct data cleaning [27], [28], [29]. As a recent representative method, CAT (Complexity, Action, and Text-spotting) is a filtering strategy which is devised to select informative image-text pairs, thereby reducing the influence of cross-modal noise [28]. In multi-spectral object detection, image registration (i.e., spatial alignment) is a commonly used pre-processing [30], [31]. By employing geometric rules to align two images, the shifted positions can be rectified across different modalities.

From the model perspective, model filtering or rectifying methods try to identify cross-modal noise samples and further remove or rectify them. To achieve this, Huang *et al.* proposed a method called Noisy Correspondence Rectifier (NCR) to tackle cross-modal noise [12]. NCR leverages the memorization effect of neural networks, dividing the data

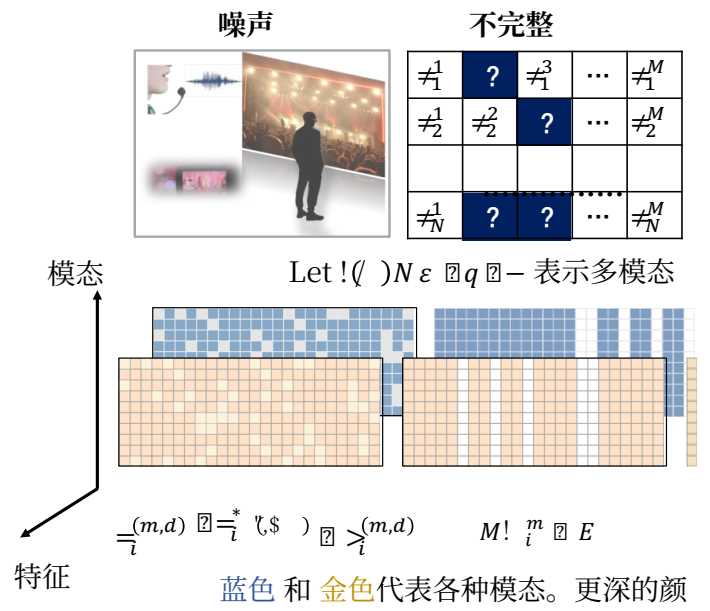


图1: 低质量多模态数据上机器学习挑战的说明。蓝色和金色代含 M 种不同模态的多模态样本, 每个模态的维度是 D . $q(x)$ 表示持程度。(a) 噪声多模态数据的质量受意外环境因素随机影响。(用信息)。(c) 不平衡多模态数据中不同模态的预期质量不同。(

正确对齐的多模态训练数据。然而, 现实世界中的多模态对往往包含弱对齐甚至不对齐的样本 [25]。例如, 在RGB-热成像多模态目标检测中, 多模态输入通常弱对齐, 即同一目标在不同模态中的位置可能发生偏移 [26]。在社交媒体上, 文本描述与图像内容经常不相关, 即不对齐的模态。在本节中, 我们将弱对齐或不对齐的多模态样本视为跨模态噪声。与模态特定噪声相比, 跨模态噪声存在于更高层次的语义空间。当前跨模态噪声减少方法大致可分为基于规则的过滤、基于模型的校正和噪声鲁棒性正则化。

从数据角度来看, 使用一些严格规则进行数据清洗 [27], [28], [29]。作为一种代表性方法, CAT (复杂度、动作和文本定位) 是一种过滤策略, 旨在选择信息丰富的图像-文本对, 从而减少跨模态噪声 [28] 的影响。在多光谱目标检测中, 图像配准 (即空间对齐) 是一种常用的预处理 [30], [31]。通过使用几何规则对齐两张图像, 不同模态之间的错位位置可以被校正。

从模型角度来看, 模型过滤或校正方法试图识别跨模态噪声样本, 并进一步移除或校正它们。为此, Huang 等人提出了名为噪声对应校正器 (NCR) 的方法来处理跨模态噪声 [12]。NCR 利用神经网络的记忆效应, 将数据

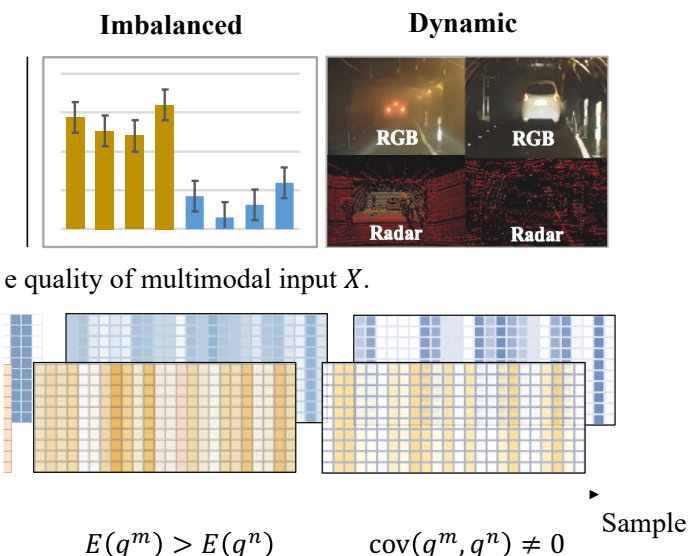


Figure 1: Quality of multimodal input X .

ities. Deeper color denotes higher quality.

low-quality multimodal data. Blue and gold represent various modes we have N multimodal samples consisting M different modalities. The quality of multimodal input, i.e., the information content of noisy multimodal data is randomly influenced by the quality of incomplete multimodal data are of zero quality (do not contain any information). (b) modalities are different for imbalanced multimodal data. (d) modalities are different for imbalanced multimodal data. (d) modalities are different for imbalanced multimodal data.

into clean and noisy subsets based on loss differences. Subsequently, it rectifies correspondence using an adaptive prediction model in a co-teaching manner. Applied to image-text matching as a showcase, NCR achieves superior performance in the presence of cross-modal noisy data. ALBEF [32] adopts momentum models to generate pseudo targets as additional supervision. BLIP [33] introduces a filter to remove the noisy data according to the similarity of image-text pairs and then utilizes a captioner to regenerate the corresponding web texts. This filtering and captioning strategy can improve the quality of multimodal pairs, contributing to the improvement of downstream vision-language tasks.

Noise robust regularization is another perspective to mitigate the influence of cross-modal noise. To explicitly stabilize and harmonize multimodal pretraining and mitigate the influence of latent noise, NLIP [34] employs noise-adaptive regularization to avoid overfitting to noisy image-text pairs, adjusting alignment labels based on estimated noisy probabilities. Additionally, NLIP utilizes a concept-conditioned cross-modal decoder to generate synthetic captions to impute the missing object information. To mitigate the influence of cross-modal noise in vision-language task, Li *et al.* [35] proposed OSCAR that detects object tags in images and uses them as anchor points for alignment. Recent work provides both theoretical and empirical analysis to shed light on the influence of cross-modal noise in multimodal contrastive learning [36]. Furthermore, the research introduces a novel MMCL loss designed to handle unpaired multimodal samples, demonstrating enhanced robustness in empirical experiments.

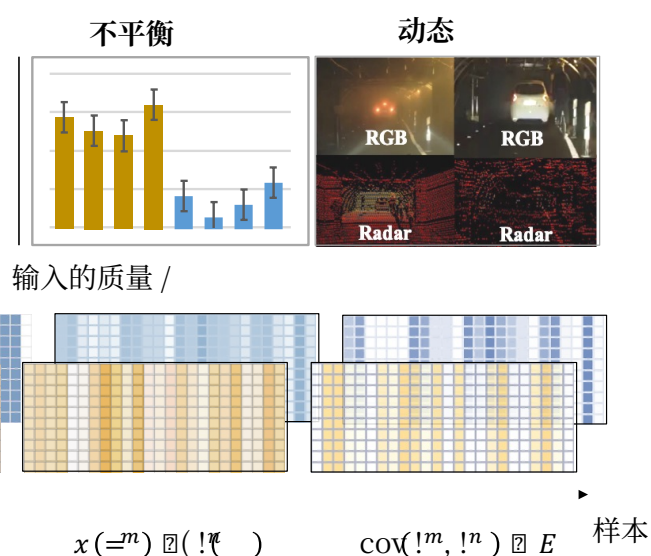


Figure 2: Quality of multimodal input X .

颜色表示更高的质量。

代表不同的模态。颜色越深表示质量越高。假设我们有 N 个包表示多模态输入的质量，即从 x 中收集的信息对下游任务的贡献。 (b) 不完全多模态数据中某些模态的质量为零（不传递任何信息）。 (d) 不同样本中不同模态的质量不同。

从模型角度来看，模型过滤或校正方法试图识别跨模态噪声样本，并进一步移除或校正它们。为此，Huang 等人提出了名为噪声对应校正器（NCR）的方法来处理跨模态噪声。NCR 利用神经网络的记忆效应，根据损失差异将数据分为干净和噪声两个子集。随后，它采用协同教学方式中的自适应预测模型来校正对应关系。以图像-文本匹配为例进行展示时，NCR 在存在跨模态噪声数据的情况下实现了优异的性能。ALBEF [32] 采用动量模型生成伪目标作为额外的监督。BLIP [33] 引入一个过滤器根据图像-文本对的相似性移除噪声数据，然后利用一个标题生成器重新生成相应的网络文本。这种过滤和标题生成策略可以提高多模态对的质量，有助于提升下游视觉语言任务的性能。

噪声鲁棒正则化是从另一个角度减轻跨模态噪声影响的方法。为了显式地稳定和协调多模态预训练并减轻潜在噪声的影响，NLIP [34] 采用噪声自适应正则化，以避免过度拟合噪声图像-文本对，并根据估计的噪声概率调整对齐标签。此外，NLIP 利用概念条件跨模态解码器生成合成字幕，以补充全缺失的物体信息。为了减轻视觉语言任务中跨模态噪声的影响，Li 等。 [35] 提出了 OSCAR，该模型检测图像中的物体标签并将其用作对齐的锚点。最近的研究提供了理论和实证分析，以阐明跨模态噪声在多模态对比学习 [36] 中的影响。此外，该研究引入了一种新的 MMCL 损失，用于处理无配对的跨模态样本，并在实证实验中展示了增强的鲁棒性。

2.3 Discussion

To summarize, learning from noisy multi-modal data is a common but challenging problem. Current approaches address this problem from two perspectives: modal-specific noise reduction (for feature noise) and cross-modal noise reduction (for semantic noise). However, these methods typically focus on specific scenarios, such as multimodal image fusion or automatic driving, leaving the exploration of general noise patterns and learning paradigms relatively unexplored. We have identified several potential research topics in this field. First, it is important to leverage the correlation between noises in different modalities. For instance, images with similar wavelengths in a hyper-spectral image often exhibit similar noise patterns. Second, it will be effective to leverage the complementarity between noisy and clean modalities for noise reduction. Third, addressing high-level semantic noise poses an interesting direction and it is more challenging. For example, how can we tackle this problem using Multimodal Large Language Models (MLLM)?

3 INCOMPLETE MULTIMODAL LEARNING

The collected multimodal data in real applications are often incomplete with partial missing modalities for some samples due to unexpected factors, such as equipment damage, data transmission and storage loss. For a concrete example, in recommendation system, the browsing behavior history and credit score information may not always be available for some users [38]. Similarly, multimodal data can be highly incomplete in medical analysis due to the patients' personal preferences, economic constraints and limitation in medical resources. For example, although combining data from multiple modalities e.g., magnetic resonance imaging (MRI) scans, positron emission tomography (PET) scans, and cerebrospinal fluid (CSF) information can obtain a more accurate diagnosis for Alzheimer's Disease [39], [40], owing to high measurement cost for PET scans and uncomfortable invasive tests for CSF, some patients may refuse to conduct these examinations. Thus, incomplete multimodal data commonly exist in Alzheimer's Disease diagnosis [41]. Typically, conventional multimodal learning models assume the completeness of multimodal data and thus cannot be applied to partial missing modalities directly. For this issue, incomplete multimodal learning, aiming at exploring information of incomplete multimodal data with partial missing modalities, appears and has received increasing research interests in recent years [42]. In this section, we mainly focus on the current progress of researches on incomplete multimodal learning. From the view of whether to impute the missing data or not, we categorize existing methods into two groups including imputation-based and imputation-free incomplete multimodal learning, where the imputation based methods are further divided into two groups as shown in Fig. 2 including instance and modality level imputation.

3.1 Imputation based incomplete multimodal learning

For the incomplete multimodal data, a natural approach to address the missing modality learning issue is to directly impute the missing modalities or related missing information

2.3 讨论

总而言之, 从噪声多模态数据中学习是一个普遍但具有挑战性的问题。当前方法从两个角度解决这个问题: 模态特定的噪声减少(针对特征噪声)和跨模态噪声减少(针对语义噪声)。然而, 这些方法通常专注于特定场景, 例如多模态图像融合或自动驾驶, 使得对通用噪声模式和学习探索相对较少。我们在该领域已经识别出几个潜在的研究主题。首先, 利用不同模态之间噪声的相关性非常重要。例如, 在超光谱图像中具有相似波长的图像通常表现出相似的噪声模式。其次, 利用噪声和干净模态之间的互补性进行噪声减少将非常有效。第三, 解决高级语义噪声是一个有趣的方向, 并且更具挑战性。例如, 我们如何利用多模态大语言模型(MLLM)来解决这个问题?

3 不完全多模态学习

在实际应用中收集到的多模态数据常常是不完整的, 由于设备损坏、数据传输和存储丢失等意外因素, 部分样本的某些模态会缺失。以具体例子为例, 在推荐系统中, 某些用户的浏览行为历史和信用评分信息可能并非总是可用 [38]。类似地, 由于患者个人偏好、经济限制和医疗资源限制, 多模态数据在医学分析中可能高度不完整。例如, 尽管结合多种模态的数据(例如, 磁共振成像(MRI)扫描、正电子发射断层扫描(PET)扫描和脑脊液(CSF)信息)可以获得更准确的阿尔茨海默病诊断 [39][40], 但由于PET扫描的高测量成本和CSF的侵入性不适检查, 一些患者可能拒绝进行这些检查。因此, 不完全多模态数据在阿尔茨海默病诊断中很常见 [41]。通常, 传统的多模态学习模型假设多模态数据的完整性, 因此不能直接应用于部分缺失的模态。针对这个问题, 不完全多模态学习旨在探索具有部分缺失模态的不完全多模态数据中的信息, 近年来已获得越来越多的研究兴趣 [42]。在本节中, 我们主要关注不完全多模态学习研究的当前进展。从是否插补缺失数据的角度来看, 我们将现有方法分为两类: 基于插补的不完全多模态学习和无插补的不完全多模态学习, 其中基于插补的方法进一步分为两类, 如图2所示, 包括实例级和模态级插补。

3.1 基于不完全多模态学习的插补

对于不完全多模态数据, 一种自然的解决缺失模态学习问题的方法是直接插补缺失的模态或相关缺失信息

in the kernels or graphs constructed from the data as shown in Fig. 2. Then the conventional multimodal learning algorithm can be applied to downstream tasks. In this branch, the existing imputation-based works can be further divided into two categories, including model-agnostic imputation and model-specific imputation. The model-specific imputation methods generally design a imputation model or network for missing data recovery while the imputation model is not necessary for the model-agnostic imputation methods which usually utilizes heuristic filling strategy for the missing modalities.

3.1.1 Model-agnostic imputation

For the missing modalities, the simplest imputation schemes are zero imputation and mean value imputation, i.e., filling missing values with zero or imputation with the averaged value in the corresponding modality, which is widely used as the baseline methods in most studies [43], [44]. However, previous work [43] shows that the naive imputation may lead to unpromising performance. To address this issue, a weighted matrix factorization model is proposed to assign small weights for the imputed modalities, mitigating the negative influence on optimization [43]. Furthermore, for an online multimodal clustering case, a dynamic weighting and imputation mechanism is designed according to the valid modalities and task at hand [45]. To implement the graph-based approach on incomplete multimodal data, Wang *et al.* [46] utilize similarity values computed from valid modalities to impute the missing elements of graph. In [47], the missing elements in the incomplete kernels are pre-imputed by the average of all columns. In [48], the authors also utilize the average instance of the corresponding modality to impute the missing instances and then assign a small weight to those imputed instances during the graph construction phase.

3.1.2 Learning based imputation

For the learning based imputation, according to the filling values, we can further divide it into two categories: kernel/graph-based imputation and raw data imputation.

Kernel/graph-based imputation: Kernel and graph learning based approaches are two main lines of incomplete multimodal learning, which transform the learning phase from the raw data to the kernel space or graph. For the kernel and graph constructed from each modality, each element represents the relationship in kernel space or affinity/similarity of instance pairs. The partial missing instances in each modality result in incomplete kernels and graphs. A line of works has been proposed to address the incomplete kernel/graph-based multimodal learning issue. Here we highlight a few seminal works. David Williams *et al.* [49] proposed a Gaussian mixture model based method that can analytically compute the missing Gaussian kernel elements associated with the missing instances. Trivedi *et al.* [50] proposed a kernelized canonical correlation analysis (KCCA) based imputation method, which minimizes a Laplacian kernel regularization to obtain the missing kernel elements. Although effective, it is only applicable to the two-modal data with one complete modality. Lately, collective kernel learning (CKL) [38] is proposed to mutually complete the two incomplete kernels associated with the two incomplete

在由数据构建的核或图中（如图2所示）。然后可以将传统的多模态学习算法应用于下游任务。在这个分支中，现有的基于插补的工作可以进一步分为两类：模型无关插补和模型相关插补。模型相关插补方法通常设计一个插补模型或网络用于缺失数据恢复，而模型无关插补方法则不需要插补模型，通常利用启发式填充策略来处理缺失模态。

3.1.1 模型无关插补

对于缺失的模态，最简单的插补方案是零插补和均值插补，即用零或对应模态中的平均值填充缺失值，这被广泛用作大多数研究中的基线方法 [43], [44]。然而，先前工作 [43] 表明朴素插补可能导致性能不佳。为解决这个问题，提出了一种加权矩阵分解模型，为插补的模态分配较小的权重，减轻其对优化 [43] 的负面影响。此外，对于在线多模态聚类情况，根据有效的模态和具体任务设计了动态加权和插补机制 [45]。为在完全多模态数据上实现基于图的方法，Wang等。 [46] 利用从有效模态计算出的相似度值来插补图中缺失的元素。在 [47] 中，不完整核中的缺失元素由所有列的平均值预先插补。在 [48] 中，作者还利用对应模态的平均实例来插补缺失的实例，然后在图构建阶段为这些插补的实例分配较小的权重。

3.1.2 基于学习的插补

对于基于学习的插补，根据填充值，我们可以将其进一步分为两类：基于核/图的插补和原始数据插补。

核/图方法：核方法和图学习方法是多模态学习中的两条主要思路，它们将学习阶段从原始数据转换到核空间或图中。对于从每个模态构建的核和图，每个元素代表核空间中的关系或实例对之间的亲和力/相似性。每个模态中的部分缺失实例导致不完整的核和图。已经提出了一系列工作来解决不完整核/图方法的多模态学习问题。在这里，我们重点介绍几项开创性工作。David Williams 等人 [49] 提出了一种基于高斯混合模型的方法，可以解析地计算与缺失实例相关的缺失高斯核元素。Trivedi 等人 [50] 提出了一种基于核化典型相关分析 (KCCA) 的插补方法，该方法最小化拉普拉斯核正则化以获得缺失的核元素。尽管有效，但它仅适用于具有一个完整模态的两个模态数据。最近，集体核学习 (CKL) [38] 被提出以相互补全与两个不完整模态相关的两个不完整核。对于具有两个以上模态的数据，例如具有模态 () 的情况，CKL 需要根据核 [38] 在一个循环迭代方案中恢复第 v 个模态的每个核矩阵。Sahely 等人提出了一种基于稀疏重建技术的多模态核补全方法。该方法可以在内核保留和内核间探索框架中联合补全所有核的缺失行和列。Yang 等人将不完整多模态插补问题简化为半监督学习问题。他们进一步提出了基于矩阵补全的 SLIM (不完整模态的半监督学习) 及其核视觉 SLIM-K，用于解决模态不完整性问题。最近，Liu 等人提出了几种灵活的核方法，用于恢复不完整多模态聚类任务中的缺失核元素。

TABLE 1: A summary of representative

Noise type	Reference	Denoise Par
Modal-specific	[20]	Joint variat
	[37]	
	[21]	
	[22]	
	[24]	
	[23]	
Cross-modal	[27], [28], [29]	Rules-based fi
	[12]	Model-based re
	[32], [33]	Noise robust reg
	[34], [36]	
	[35]	

噪声类型	参考文献	降噪范式
模态特定	[20]	联合变化
	[37]	
	[21]	
	[22]	
	[24]	
	[23]	
跨模态	[27], [28], [29]	基于规则的过
	[12]	基于模型的校
	[32], [33]	噪声鲁棒正则化
	[34], [36]	
	[35]	

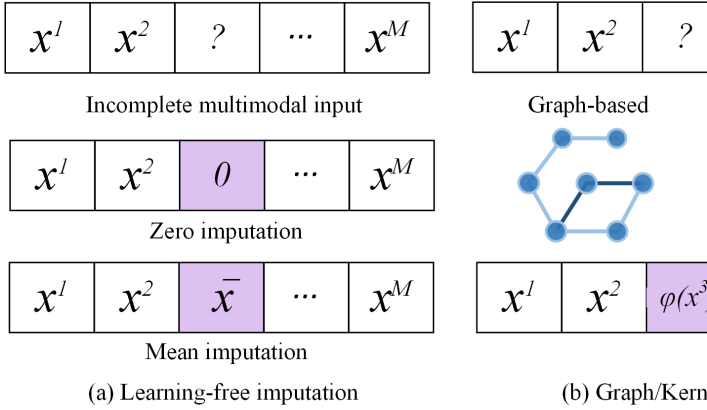


Fig. 2: Imputation based inco

modalities. For the data with more than two modalities, such as the case with k modalities ($k \geq 3$), CKL needs to recover each kernel matrix K_{i+1} of the $(i+1)$ th modality according to kernel K_i in a cycle iteration scheme. Sahely *et al.* [51] proposed a multimodal kernel completion method based on the sparse reconstruction technique. The method can complete the missing rows and columns of all kernels jointly in a within-kernel preservation and between-kernel exploration framework. Yang *et al.* [52] reduced incomplete multi-modal imputation into semi-supervised learning problem. They further proposed SLIM (Semi-supervised Learning with Incomplete Modality) and its kernel vision SLIM-K based on matrix completion for solving the modal incompleteness. Recently, Liu *et al.* proposed several flexible kernel methods to recover the missing kernel elements for incomplete multimodal clustering tasks [53], [54], [55], [56].

By aligning all incomplete kernels to a consensus kernel derived from the consensus orthogonal representation, these methods can jointly obtain the missing elements associated with missing instances and the consensus partition representation shared by all modalities for incomplete multimodal data clustering. Besides of kernel-based methods, an adaptive graph completion method is developed to recover the missing similarities of all graphs with respect to all modalities [57]. The method mainly employs the sparse-representation-based data reconstruction technique to borrow the graph information of other modalities in recovering the

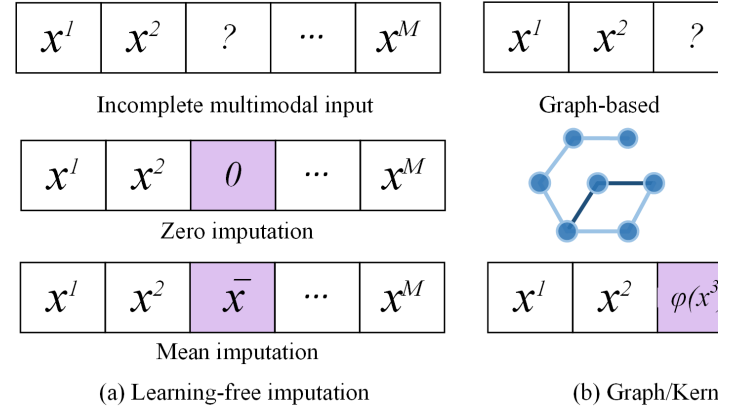


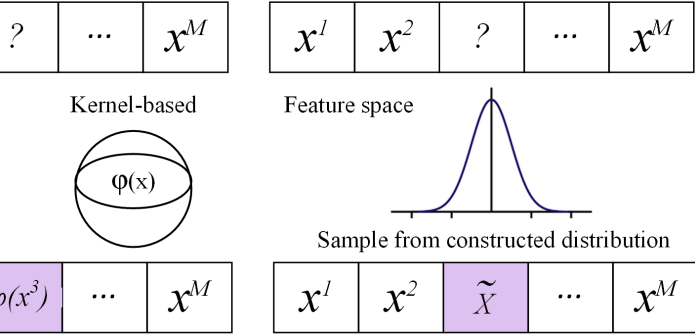
图2: 基于插补的不完整多模态

对于具有两个以上模态的数据, 例如具有 k 模态 ($k \geq 3$) 的情况, CKL需要根据核 K_i 在一个循环迭代方案中恢复第 $\langle v6 \rangle$ 个模态的每个核矩阵 K_{i+1} 。Sahely 等人. [51] 提出了一种基于稀疏重建技术的多模态核补全方法。该方法可以在内核保留和内核间探索框架中联合补全所有核的缺失行和列。Yang 等人. [52] 将不完整多模态插补问题简化为半监督学习问题。他们进一步提出了基于矩阵补全的SLIM (不完整模态的半监督学习) 及其核视觉SLIM-K, 用于解决模态不完整性问题。最近, Liu 等人提出了几种灵活的核方法, 用于恢复不完整多模态聚类任务 [53], [54], [55], [56]中的缺失核元素。

通过将所有不完整的核对齐到一个从共识正交表示中推导出的共识核, 这些方法可以联合获得与缺失实例相关的缺失元素以及所有模态共享的用于不完整多模态数据聚类的共识划分表示。除了基于核的方法之外, 还开发了一种自适应图补全方法来恢复所有模态下所有图的相关缺失相似度 [57]。该方法主要采用基于稀疏表示的数据重建技术来借用其他模态的图信息以进行恢复。

ive multi-modal noise reduction works.

aradigm	Application
riation	Image fusion
	Remote-sensing scene classification
d filtering	Multimodal pretraining
d rectifying	Image-text matching
egularization	Multimodal pretraining
	Vision-language understanding



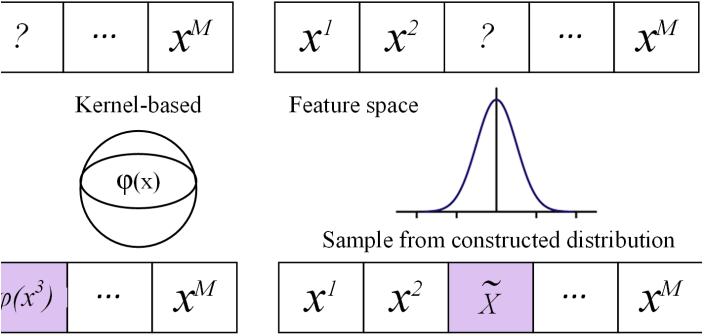
Kernel imputation (c) Raw data imputation

omplete multimodal learning.

, missing elements of each modality.

Raw data imputation: Some methods recover the missing modalities at raw feature level. For example, VIGAN (View Imputation via Generative Adversarial Network) [58] is a pioneering work based on generative adversarial network (GAN) and Autoencoder for missing modality recovery. VIGAN employs GAN to initialize the missing modality according to another modality of the same sample and then uses denoise Autoencoder to refine the recovered missing modalities. A limitation of VIGAN is that it is not applicable to data with more than two modalities. Subsequently, partial multimodal clustering via consistent GAN (PMVC_CGAN) [59] and adversarial incomplete multimodal clustering (AIMC) [60] are proposed for missing modality imputation and incomplete multimodal clustering. Different from VIGAN which uses one available data to generate the corresponding missing modality, PMVC_CGAN and AIMC seek to learn the missing modalities from the common representation encoded by the un-missing modalities of the same sample. In [61], a two-stage incomplete deep multimodal network is proposed for breast cancer prediction. Similar to the network structures of AIMC and PMVC_CGAN, GAN is introduced to enhance the quality of missing view imputation. Besides of GAN-based methods, CRA (cascaded residual autoencoder) stacks all modalities into one modal and treats the missing modalities imputation as conventional missing feature completion by optimizing a single-

	应用
	图像融合
	遥感场景分类
过滤	多模态预训练
	图像-文本匹配
正	多模态预训练
	多模态预训练
	视觉-语言理解



Kernel imputation (c) Raw data imputation

态学习。

每种类别的缺失元素。

原始数据插补: 一些方法在原始特征层上恢复缺失的模态。例如, VIGAN (基于生成对抗网络 (GAN) 和自动编码器的视图插补) [58] 是一项开创性工作, 用于缺失模态恢复。VIGAN采用GAN根据同一样本的另一个模态初始化缺失模态然后使用去噪自动编码器来细化恢复的缺失模态。VIGAN的一个局限性是它不适用于具有两个以上模态的数据。随后, 通过一致性GAN的部分多模态聚类 (PMVC CGAN) [59] 和对抗性不完整多模态聚类 (AIMC) [60]被提出用于缺失模态插补和不完整多模态聚类。与VIGAN使用一个可用数据来生成相应的缺失模态不同, PMVC CGAN和AIMC试图从同一样本的未缺失模态编码的共同表示中学习缺失模态。在 [61],中, 提出了一种用于乳腺癌预测的两阶段不完整深度多模态网络。与AIMC和PMVC CGAN的网络结构类似, 引入了GAN来提高缺失视图插补的质量。除了基于GAN的方法外 CRA (级联残差自动编码器) 将所有模态堆叠到一个模态中, 并将缺失模态插补视为通过优化单个-

modal cascaded residual autoencoder network [62]. Liu *et al.* [63] proposed another representative autoencoder based imputation framework termed RecFormer for clustering task. Specifically, RecFormer employs a two-stage autoencoder network with the self-attention structure to synchronously extract high-level semantic representations and recover the missing data

To obtain a more reliable modality imputation performance, Tang *et al.* [64] proposed to impute the modality as the average of corresponding semantic neighbors. Lin *et al.* [65] established a dual prediction model that can predict the latent representation of the missing modality according to that of the observed modality and then used the decoder to obtain the recovered missing modality according to the predicted latent representation. matrix factorization is also applied for raw modality imputation. For example, UEAF (Unified Embedding Alignment Framework) [66] and IMCRV (Incomplete multimodal Clustering with Reconstructed Views) [67] provide two matrix factorization based missing modality recovery models, which can reversely reconstruct the data of missing modalities from the common representation. In [68], low-rank representation and tensor learning are introduced for joint missing modality recovery and graph completion. Explicitly taking the reliability of imputation into account, Xie *et al.* [69] introduced the Uncertainty-induced Incomplete Multi-View Data Classification (UIMC) model to address challenges in classifying incomplete multi-view data with missing perspectives. UIMC explores and exploits uncertainty by imputing missing modalities as distributions rather than deterministic values, allowing for more reliable predictions. The model employs an evidential multimodal learning framework to weigh imputed data, reducing the impact of low-quality imputations. In incomplete multimodal action recognition task, Sangmin *et al.* proposed a modular network called ActionMAE, which randomly drops modality features and then learn to impute them.

Lin *et al.* proposed a novel objective by integrating representation learning and data recovery into a unified information-theoretic framework termed COMPLETER [70] for clustering task. Specifically, COMPLETER propose to learn intra-modal consistency by maximizing mutual information and learn modal imputation by minimizing conditional entropy. Dong *et al.* proposed a deep Gaussian processes model to impute the missing modalities and further enhance the performance of downstream tasks [71]. Motivated by contrastive learning, Yang *et al.* introduced a multimodal imputation paradigm termed SURE for robust multimodal imputation [72]. The method treats complete multimodal pairs as positive and randomly sampling unpaired samples as negative. A noise-robust contrastive loss is employed to alleviate the influence of false negatives. Li *et al.* proposed a prototype-based multimodal imputation method, which incorporates a two-stream model with dual attention layers and contrastive learning mechanism to learn modal-specific prototypes and model the cross-modal relationship [73]. Considering incomplete multimodal inputs, the model performs modality imputation using prototypes learned from the incomplete multimodal training data and the sample-prototype relationship inherited from the observed modalities. Xu *et al.* [74] proposed a novel federated multi-view clustering method using an unsupervised tech-

模态级联残差自动编码器网络 [62]。刘等。[63] 为聚类任务提出了一种基于自动编码器的代表性插补框架，称为 RecFormer。具体而言，RecFormer采用具有自注意力结构的两阶段自动编码器网络，以同步提取高级语义表示并恢复缺失数据

为获得更可靠的模态插补性能，Tang 等。[64]提出将模态插补为对应语义邻域的平均值。Lin 等。[65]建立了一个双预测模型，该模型可以根据观测模态的潜在表示来预测缺失模态的潜在表示，然后使用解码器根据预测的潜在表示获得恢复的缺失模态。矩阵分解也应用于原始模态插补。例如，UEAF（统一嵌入对齐框架）[66]和IMCRV（不完整多模态聚类与重建视图）[67]提供了两种基于矩阵分解的缺失模态恢复模型，这些模型可以从公共表示中逆向重建缺失模态的数据。在 [68] 中，引入了低秩表示和张量学习，用于联合缺失模态恢复和图补全。明确考虑插补的可靠性，Xie等。

[69] 引入了不确定性诱导的不完整多视图数据分类（UIMC）模型来解决分类具有缺失视角的不完整多视图数据的挑战。UIMC通过将缺失模态插补为分布而不是确定性值来探索和利用不确定性，从而允许更可靠的预测。该模型采用证据多模态学习框架来权衡插补数据，减少低质量插补的影响。在不完整多模态动作识别任务中，Sangmin 等。提出了一个名为ActionMAE的模块化网络，该网络随机丢弃模态特征，然后学习对其进行插补。

Lin 等提出了一个新颖的目标，通过将表征学习和数据恢复集成到一个统一的信息论框架COMPLETER [70]中，用于聚类任务。具体来说，COMPLETER提出通过最大化互信息来学习模态内一致性，并通过最小化条件熵来学习模态插补。Dong等提出了一个深度高斯过程模型来插补缺失的模态，并进一步提升了下游任务的性能 [71]。受对比学习的启发，Yang等引入了一种名为SURE的多模态插补范式，用于鲁棒的多模态插补 [72]。该方法将完整的多模态对视为正例，并随机采样无对例作为负例。采用了一种抗噪声对比损失来减轻假负例的影响。Li等提出了一种基于原型的多模态插补方法，该方法结合了一个双流模型、双重注意力层和对比学习机制，用于学习模态特定的原型并建模跨模态关系 [73]。考虑到不完整的多模态输入，该模型使用从不完整的多模态训练数据中学习到的原型以及从观察到的模态继承的样本-原型关系来执行模态插补。Xu 等。[74] 提出了一种使用一种独特的用于评估和改进插补质量的无监督技术的新型联邦多视图聚类方法，能够高效处理不完整多视图数据的各种场景。

nique to evaluate and refine imputation quality, efficiently handling various scenarios of incomplete multi-view data.

3.2 Imputation-free incomplete multimodal learning

Different from the above imputation based methods, imputation-free methods only focus on exploring the information corresponding to the available modalities. Most of the related imputation-free methods can be grouped into four mainstreams: latent representation and projection learning, graph learning, kernel learning, and deep learning.

3.2.1 Latent representation and projection learning

Latent representation and projection learning based incomplete multimodal learning (taking clustering for example) generally seek to obtain a multimodal latent representation by exploring the available partially aligned information among these available modalities. In this branch, partial multimodal clustering (PMVC) [75] is a popular and pioneering work based on matrix factorization. It concentrates on encoding the data with two fully observed modalities into a common representation within the latent subspace. Based on PMVC, incomplete multimodal grouping (IMG) [44] and partial multimodal subspace clustering [76] further introduce the graph constraints to capture the structure information in the latent common subspace. A basic model used in the above methods to learn the latent common representation from the incomplete bi-modal data.

Zhou *et al.* [77] integrated the latent representation learning and regression into a unified framework for Alzheimer's disease diagnosis based on incomplete multimodal neuroimaging and genetic data. However, a widely recognized limitation of the above methods based on model (4) is that they are only applicable to the case with bi-modal data. To address this limitation, weighted matrix factorization is introduced, where doubly aligned incomplete multimodal clustering [78], one-pass incomplete multimodal clustering [79], and localized sparse incomplete multimodal clustering [80] are representative methods. These models generally impose the location information of missing instances as the weighted matrices with binary values on the matrix factorization model to eliminate the loss yet reduce the negative influence of the missing instances. The basic model of the weighted matrix factorization based methods can be formulated as follows:

$$\min_{U^{(m)}, P} \sum_{m=1}^M \left\| W^{(m)} \left(X^{(m)} - PU^{(m)} \right) \right\|_F^2 + \Psi \left(U^{(m)}, P \right), \quad (3)$$

where $X^{(m)}$ denotes the data of the m -th modality with missing instances being 0 or the average value of the available instances in the modality. $U^{(m)}$ denotes the basic matrix and $\Psi \left(U^{(m)}, P \right)$ denotes the constraints of these variables. P is the latent complete common representation to learn. $W^{(m)}$ is a diagonal matrix constructed according to the instance missing location information in the m -th modality. For example, if the i -th sample does not have the m -th modality, then $W_{i,i}^{(m)}$ can be simply set as 0, otherwise $W_{i,i}^{(m)} = 1$. With such a setting, the negative influence of missing modalities can be eliminated from the learning model, and the available aligned consistent and

nique来评估和改进插补质量, 高效处理不完整多视图数据的各种场景。

3.2 无插补的不完整多模态学习

与上述基于插补的方法不同, 无插补方法仅关注探索与可用模态相对应的信息。大多数相关无插补方法可分为四大主流: 潜在表示与投影学习、图学习、核学习以及深度学习。

3.2.1 潜在表示与投影学习

基于潜在表示与投影学习的不完整多模态学习 (以聚类为例) 通常通过探索这些可用模态之间部分对齐的信息来获取多模态潜在表示。在这一分支中, 部分多模态聚类 (PMVC) [75] 是基于矩阵分解的一种流行且开创性的工作。它专注于将带有两个完全观测模态的数据编码到潜在子空间的共同表示中。基于PMVC, 不完整多模态分组 (IMG) [44] 和部分多模态子空间聚类 [76] 进一步引入图约束以捕获潜在共同子空间中的结构信息。上述方法中用于从不完整双模态数据学习潜在共同表示的基本模型。

周 等。 [77] 将潜在表示学习和回归整合到一个统一的框架中, 用于基于不完整的多模态神经影像和遗传数据的阿尔茨海默病诊断。然而, 基于模型 (4) 的上述方法的一个被广泛认可的局限性是, 它们仅适用于双模态数据的情况。为了解决这个问题, 引入了加权矩阵分解, 其中双重对齐的不完整多模态聚类 [78], 单次不完整多模态聚类 [79], 和局部稀疏不完整多模态聚类 [80] 是典型方法。这些模型通常将缺失实例的位置信息作为加权矩阵分解模型上的二值权重矩阵, 以消除损失并减少缺失实例的负面影响。基于加权矩阵分解的方法的基本模型可以表述如下:

$$\min_{U^{(m)}, P} \sum_{m=1}^M \left\| W^{(m)} \left(X^{(m)} - PU^{(m)} \right) \right\|_F^2 + \Psi \left(U^{(m)}, P \right), \quad (3)$$

$X^{(m)}$ 表示第 m 种模态的数据, 其中缺失实例为0或该模态中可用实例的平均值。 $U^{(m)}$ 表示基本矩阵, $\Psi \left(U^{(m)}, P \right)$ 表示这些变量的约束。 P 是用于学习的潜在完整公共表示。 $W^{(m)}$ 是根据第 m 种模态中实例缺失位置信息构造的对角矩阵。例如, 如果第 i 个样本没有第 m 种模态, 则 $W_{i,i}^{(m)}$ 可以简单地设置为0, 否则 $W_{i,i}^{(m)} = 1$ 。通过这种设置, 缺失模态的负面影响可以从学习模型中消除, 并且可以探索可用对齐一致的和

complementary information can be explored for the latent common representation learning.

Besides of these weighted matrix factorization models, a generalized incomplete multimodal clustering model is proposed to learn a complete common representation from the incomplete modal-specific representations with a partially aligned learning term [81]. Specifically, such a partially aligned learning term can be regarded as a variant of the weighted common representation learning. To prevent the model from overly focusing on certain modalities with larger dimensions, projective incomplete multimodal clustering transforms the matrix factorization into a variant of the weighted projection learning model to obtain the consensus representation from the incomplete multimodal data [82]. For the incomplete multimodal and incomplete multi-label classification task, a variant of the weighted projection learning model [83] and weighted matrix factorization model [84] are developed to explore the information of available modalities. In terms of infant neuroscience, Zhang *et al.* propose to explore incomplete multimodal data by constructing a latent representation to capture the complementary information from incomplete time-points [85]. In summary, the above methods focus on exploring the available information of the observed modalities by introducing some techniques like the weighting mechanism to eliminate the loss of those missing modalities.

3.2.2 Graph learning based

Graph learning based method is very popular in conventional multimodal clustering. This technique is also widely considered and integrated into other methods like matrix factorization based and deep learning based methods, to explore the structure information of data so as to enhance the performance. For incomplete multimodal learning, the main challenges to apply the graph learning based approach are two-folds: (1) How to obtain the intrinsic structure relationships among the available instances in each modality? (2) How to fully explore these incomplete relationships? For example, for the incomplete multimodal clustering tasks, one of the major challenges is to obtain the consensus complete graph or representation from these incomplete modal-specific relationships associated with all incomplete modalities. Specifically, we can formulate this issue as the following simple framework:

$$\min_{\{S^{(m)}\}_{m=1}^l, Z} \sum_{m=1}^l f(\bar{X}^{(m)}, S^{(m)}) + g(S^{(m)}, W^{(m)}, Z), \quad (4)$$

where $\bar{X}^{(m)}$ denotes the set of the available instances in the m -th modality. $S^{(m)}$ represents the modal-specific graph learned from the m -th modal data $\bar{X}^{(m)}$. $W^{(m)}$ is the modal-available or missing indicator matrix as defined in Eq. 3. Taking the incomplete multimodal clustering task for example, the target of general graph learning based method is to learn the consensus graph or representation Z from the incomplete data $\{\bar{X}^{(m)}\}_{m=1}^l$ or graph $\{S^{(m)}\}_{m=1}^l$, such that a consensus clustering result can be obtained by performing k-means or spectral clustering on Z .

As one representative work, incomplete multimodal spectral clustering [86] proposed a unified framework that

互补信息可以被探索用于潜在公共表示学习。

除了这些加权矩阵分解模型，还提出了一个广义不完全多模态聚类模型，该模型通过带有部分对齐学习项 [81] 的不完全模态特定表示来学习完整的共同表示。具体来说，这种部分对齐学习项可以被视为加权共同表示学习的一种变体。为了防止模型过度关注维度较大的某些模态，投影不完全多模态聚类将矩阵分解转化为加权投影学习模型的一种变体，以从不完全多模态数据 [82] 中获得共识表示。对于不完全多模态和不完全多标签分类任务，开发了一种加权投影学习模型 [83] 和加权矩阵分解模型 [84]，以探索可用模态的信息。在婴儿神经科学方面，Zhang等人提出通过构建潜在表示来探索不完全多模态数据，以从不完全时间点 [85] 中捕获互补信息。总之，上述方法通过引入一些技术（如加权机制）来探索观察到的模态的可用信息，以消除缺失模态的损失。

3.2.2 基于图学习的方法

基于图学习的方法在传统的多模态聚类中非常流行。这种技术也被广泛考虑并集成到其他方法中，如基于矩阵分解的方法和基于深度学习的方法，以探索数据的结构信息从而提升性能。对于不完整的多模态学习，应用基于图学习的方法的主要挑战有两个：(1) 如何获取每个模态中可用实例之间的内在结构关系？(2) 如何充分探索这些不完整的关系？例如，对于不完整的多模态聚类任务，一个主要挑战是从与所有不完整模态相关的这些不完整模态特定关系中获取共识完整图或表示。具体来说，我们可以将这个问题表述为以下简单的框架：

$$\min_{\{S^{(m)}\}_{m=1}^l, Z} \sum_{m=1}^l f(\bar{X}^{(m)}, S^{(m)}) + g(S^{(m)}, W^{(m)}, Z), \quad (4)$$

其中 $\bar{X}^{(m)}$ 表示第 m 个模态中可用实例的集合。 $S^{(m)}$ 表示从第 m 个模态数据 $\bar{X}^{(m)}$ 中学习到的模态特定图。

$W^{(m)}$ 是根据公式 3 定义的模态可用或缺失指示矩阵。以不完整的多模态聚类任务为例，基于图学习方法的目的是从不完整数据 $\{\bar{X}^{(m)}\}_{m=1}^l$ 或图 $\{S^{(m)}\}_{m=1}^l$ 中学习共识图或表示

Z ，以便通过在 Z 上执行 k-means 或谱聚类来获得共识聚类结果。

作为一项代表性工作，不完全多模态谱聚类 [86] 提出了一种统一的框架，该框架

integrates the adaptive collaborative representation based modal-specific graph learning and co-regularized consensus representation learning. The first term enables the method to adaptively capture the intrinsic relationships among available instances of each modality and the second term seeks to obtain the consensus complete representation by fully exploring such similarity relationship. For the data with only two modalities, Wu *et al.* [87] proposed a method based on structured graph learning, which seeks to learn two modal-specific graphs corresponding to the two modalities with a partially aligned constraint on the samples with complete modalities. Then, the method obtains the modal-specific clustering from the two graphs and adopts the best one-to-one map to obtain the final consensus clustering result. The work in [88] pre-constructs several modal-specific graphs from the available instances of every modality and then learns a consensus graph from these graphs via a weighted consensus graph learning model with the constraint of pre-constructed confident neighbor information. In summary, except for the imputation based graph methods, most graph-based incomplete multimodal learning methods seek to exploit certain similarity information among available instances to discover the clustering structure and clustering result. In [89], for Alzheimer's disease prediction, a unified linear regression based model is developed, in which the incomplete multimodal data are segmented into several complete subsets for model optimization. Similar to the above method, Liu *et al.* [90] segmented the incomplete multimodal data into several complete subsets and used the sparse representation algorithm to construct the sub-graphs from these subsets. Then a modal-aligned hypergraph learning model is developed to perform the Alzheimer's disease diagnosis.

3.2.3 Kernel learning based

Kernel learning based methods transform the learning phase from the raw-feature space into the kernel space. Similar to the graph learning based methods, each element of the kernel matrix denotes a kind of the similarity or connection values of the corresponding two samples. For incomplete multimodal data, the constructed kernels w.r.t. different modalities will also be incomplete with missing rows and columns. Thus, the target of kernel learning based methods for incomplete multimodal data is to explore the information of these incomplete kernels. Late fusion incomplete multimodal clustering [91] is one of the representative kernel imputation-free methods for incomplete multimodal data. The method does not need to impute the missing kernel elements but tries to simultaneously impute the missing representations corresponding to the missing modalities and learn the consensus representation in a joint framework. Furthermore, to reduce the low-quality imputation of the latent representations, Liu *et al.* [92] proposed an efficient and effective regularized incomplete multimodal clustering method (EERIMVC), which integrates the initialized consensus representation as prior knowledge. In this work, such prior representation is obtained by the conventional complete kernel algorithm with zero or average filling. Therefore, the quality of the initialized consensus representation will be an important factor to the performance. To break the gap between representation learning and label prediction, a one-

将基于自适应协作表示的模态特定图学习和协同正则化共识表示学习相结合。第一个术语使该方法能够自适应地捕获每个模态的可用实例之间的内在关系，第二个术语则试图通过充分探索这种相似性关系来获得共识完整表示。对于只有两种模态的数据，Wu等。[87]提出了一种基于结构化图学习的方法，该方法试图在具有完整模态的样本上施加部分对齐约束，从而学习对应于两种模态的两个模态特定图。然后，该方法从这两个图中获得模态特定聚类，并采用最佳一对一映射来获得最终共识聚类结果。在[88]中，该方法从每种模态的可用实例中预先构建几个模态特定图，然后通过一个带有预先构建的置信邻居信息约束的加权共识图学习模型从这些图中学习共识图。总之，除了基于插补的图方法外，大多数基于图的不完全多模态学习方法都试图利用可用实例之间的某些相似性信息来发现聚类结构和聚类结果。在[89]中，针对阿尔茨海默病预测，开发了一种基于线性回归的统一模型，其中不完全多模态数据被分割成几个完整的子集以进行模型优化。类似于上述方法，Liu等。[90]将不完全多模态数据分割成几个完整的子集，并使用稀疏表示算法从这些子集中构建子图。然后开发了一种模态对齐超图学习模型来执行阿尔茨海默病诊断。

3.2.3 基于核学习

基于核学习的方法将学习阶段从原始特征空间转换到核空间。类似于基于图学习的方法，核矩阵的每个元素都表示对应两个样本的一种相似性或连接值。对于不完全多模态数据，针对不同模态构建的核也会因缺失行列而不完整。因此，针对不完全多模态数据的基于核学习的方法的目标是探索这些不完全核的信息。晚期融合不完全多模态聚类[91]是不完全多模态数据的一种代表性无插补核学习方法。该方法无需插补缺失的核元素，而是尝试在联合框架中同时插补缺失模态对应的缺失表征，并学习共识表征。此外，为了减少潜在表征的低质量插补，刘等[92]提出了一种高效且有效的正则化不完全多模态聚类方法(EERIMVC)，该方法将初始化的共识表征作为先验知识进行整合。在本工作中，该先验表征是通过传统完整核算法进行零值或平均填充获得的。因此，初始化共识表征的质量将成为性能的重要影响因素。为了弥合表征学习与标签预测之间的差距，提出了一种单阶段不完全多模态聚类方法[91]。与EERIMVC相比，该方法用聚类标签矩阵和一个正交矩阵替换了共识表征。

stage incomplete multimodal clustering method is proposed [93]. Compared with EERIMVC, this method replaces the consensus representation with a clustering label matrix and an orthogonal matrix.

For the above late fusion based kernel methods, we can observe that: (1) These methods can simultaneously recover the modal-specific missing representations and learn the consensus representation by exploring the partially aligned structure information of data. (2) The performance of these methods highly relies on the quality of the initialized incomplete basis representations derived from the incomplete kernels.

3.2.4 Deep learning based

Recent years, deep learning received lots of attention in incomplete multimodal learning. For instance, Zhao *et al.* [94] proposed a deep partial multimodal clustering network, which first uses deep encoders to extract the latent features of multimodal data and then adopts the graph regularized partial multimodal clustering model to learn the consensus representation from the incomplete bi-modal data. For data with more than two modalities, Wen *et al.* [95] proposed several deep incomplete multimodal clustering networks based on deep autoencoder framework, cognitive learning [96], and graph neural network [97], respectively. The objective function of these methods can be unified as the following formula:

$$\min \sum_{m=1}^l f(X^{(m)}, \bar{X}^{(m)}) + \lambda g\left(Z, \{Z^{(m)}\}_{m=1}^l\right), \quad (5)$$

where $X^{(m)} \in R^{n \times d_m}$ and $\bar{X}^{(m)} \in R^{n \times d_m}$ denote the original and reconstructed m -th modal data. n denotes the number of samples and d_m is the feature number of the m -th modal sample. $W^{(m)} \in \{0, 1\}^{n \times n}$ is a diagonal binary matrix as defined in (3) to indicate the location of the missing modalities. $\bar{Z}$ and $\{Z^{(m)}\}_{m=1}^l$ represent the consensus or fused latent representation and the modal-specific latent representation at one layer of the network. λ is penalty parameter. f and g are functions or constraints for these variables. For example, f is set as the mean square error (MSE) loss function in many methods. Objective (5) can be also split into two independent losses to optimize two networks separately, where the first loss with the function f is generally used for network pre-training. Similar to these unsupervised multimodal learning methods, Liu *et al.* [98] developed an incomplete multimodal multi-label classification network, with the autoencoder based framework and multimodal contrastive constraint.

The above methods focus on the observed instances to train the network and can flexibly handle all kinds of incomplete data with more than two modalities by introducing the location matrix $W^{(m)}$ as prior weight in function f . In addition, a common technique adopted in these three methods is the weighted latent representation fusion, in which the location information of missing modalities is adopted as fusion weight to reduce the negative influence of missing modalities. Lately, Xu *et al.* [99] proposed an imputation-free and fusion-free deep network for incomplete multimodal data clustering. The method mines the multimodal cluster

与 EERIMVC 相比, 该方法用聚类标签矩阵和一个正交矩阵替换了共识表征。

对于上述基于晚期融合的方法, 我们可以观察到: (1) 这些方法可以同时恢复特定模态的缺失表示并学习共识表示, 通过探索数据的部分对齐结构信息。(2) 这些方法的性能高度依赖于从不完整核中推导出的初始化不完全基表示的质量。

3.2.4 基于深度学习

近年来, 深度学习在不完整多模态学习中受到了广泛关注。例如, Zhao 等人 [94]提出了一种深度部分多模态聚类网络, 该网络首先使用深度编码器提取多模态数据的潜在特征, 然后采用图正则化部分多模态聚类模型从不完整双模态数据中学习共识表示。对于具有超过两种模态的数据, Wen 等人 [95]分别基于深度自动编码器框架、认知学习 [96]和图神经网络 [97],提出了几种深度不完整多模态聚类网络。这些方法的目标函数可以统一表示为以下公式:

$$\min \sum_{m=1}^l f(X^{(m)}, \bar{X}^{(m)}) + \lambda g\left(Z, \{Z^{(m)}\}_{m=1}^l\right), \quad (5)$$

$X^{(m)} \in R^{n \times d_m}$ 和 $\bar{X}^{(m)} \in R^{n \times d_m}$ 分别表示原始和重建的 m -模态数据。 n 表示样本数量, d_m 是第 m 模态样本的特征数量。 $W^{(m)} \in \{0, 1\}^{n \times n}$ 是一个对角二进制矩阵, 如(3)中定义, 用于指示缺失模态的位置。 $\bar{Z}$ 和 $\{Z^{(m)}\}_{m=1}^l$ 分别表示网络某一层的共识或融合潜在表示以及模态特定潜在表示。 λ 是惩罚参数。 f 和 g 是这些变量的函数或约束。例如, 在许多方法中, f 被设置为均方误差 (MSE) 损失函数。目标(5)也可以拆分为两个独立的损失, 分别优化两个网络, 其中第一个使用函数 f 的损失通常用于网络预训练。类似于这些无监督多模态学习方法, Liu 等人 [98]开发了一个不完整的多模态多标签分类网络, 该网络基于自动编码器框架和多模态对比约束。

上述方法专注于观测到的实例来训练网络, 并通过在函数 f 中引入位置矩阵 $W^{(m)}$ 作为先验权重, 能够灵活处理具有两个以上模态的不完整数据。此外, 这三种方法中采用的一种常见技术是加权潜在表示融合, 其中将缺失模态的位置信息用作融合权重, 以减少缺失模态的负面影响。最近, Xu 等 [99] 提出了一种针对不完整多模态数据聚类的无插补和无融合深度网络。该方法挖掘子空间中的多模态互补性, 并将此类互补信息转换为监督信息来指导网络训练。然而, 这种方法需要一部分具有完全观测模态的样本用于模型训练。

complementarity in the subspace and transforms such complementary information as supervised information to guide the network training. However, this method needs a part of samples with fully observed modalities for model training.

For supervised learning tasks, a incomplete multimodal learning framework is proposed in [100], which can flexibly learn a common representation for arbitrary missing data and preserve both common and complementary information. For the more difficult task with incomplete labels, a deep double incomplete multimodal multi-label learning network [101] is proposed. This deep network mainly introduces two mask constraints *w.r.t.* the missing view locations and missing label locations, to reduce the negative impact of the missing labels and missing views. A good property of the method is that it can handle all kinds of incomplete data classification tasks including supervised/semi-supervised cases, complete/incomplete data, and complete/incomplete label cases. More recent work [102] focuses on an even more generalized scenario than incomplete multimodal learning which is so called *train-test modality mismatch*, where the test multimodal data can contains more or less modalities than training data. Specifically, all modalities are available at pretraining stage, while a completely different subset is encountered by the model during supervised training and testing in downstream tasks.

In medical image processing, Sanaz *et al.* proposed MMCFormer [103], a novel missing modality compensation network to address the incomplete modalities for brain tumor segmentation. The proposed method is build upon 3D efficient transformer blocks and uses a co-training strategy to effectively train a missing modality network by ensuring feature consistency between complete modalities and incomplete modalities. Yao *et al.* proposed DrFuse [104] for effective medical diagnosis and forecasting prognosis with incomplete multimodal data. DrFuse first learns the common features shared between all modalities and modality-specific features within each separated modality and then weight and then combining the prediction from each modality to make the final prediction. Specifically, a disease-wise attention layer is devised to produce the patient- and disease-wise weights to tackle the modality inconsistency issue.

Some representative incomplete multimodal learning methods are summarized in Table 4.

3.3 Discussion

To sum up, the ways to handle incomplete data can be commonly divided into two categories, imputation-based incomplete multimodal learning and imputation-free incomplete multimodal learning. Methods that focus on imputing incomplete modalities can be classified into two types: one indirectly recovers the missing modalities through graph or kernel completion, and the other directly fills in the original data. For imputation-free incomplete multimodal learning methods, their design philosophy focuses more on utilizing available modalities to explore information and understand multimodal data. However, both types of methods for handling incomplete multimodal data face deeper challenges. For example, the quality assessment of imputed instances for missing modalities is commonly overlooked. Besides, using prior missing information to

互补性, 并将其转换为监督信息来指导网络训练。然而, 这种方法需要一部分具有完全观测模态的样本用于模型训练。

针对监督学习任务, [100], 中提出了一种不完整的多模态学习框架, 该框架能够灵活地为任意缺失数据学习公共表征, 并保留公共信息与互补信息。对于更困难的带缺失标签的任务, 提出了深度双重不完整多模态多标签学习网络 [101] 该深度网络主要引入了两种掩码约束 *w.r.t.* 缺失视图位置和缺失标签位置, 以减少缺失标签和缺失视图的负面影响。该方法的一个良好特性是能够处理各种不完整数据分类任务, 包括监督/半监督情况、完整/不完整数据以及完整/不完整标签情况。更近期的相关工作 [102] 关注于比不完整多模态学习更通用的场景, 即所谓的 训练-测试模态不匹配, 其中测试多模态数据可以包含比训练数据更多或更少的模态。具体而言, 所有模态在预训练阶段都可用, 而在下游任务的监督训练和测试过程中, 模型会遇到完全不同的子集。

在医学图像处理中, Sanaz 等提出了MMCFormer [103], 这是一种新型缺失模态补偿网络, 用于解决脑肿瘤分割中的不完全模态问题。该方法基于3D高效Transformer模块构建并采用协同训练策略, 通过确保完整模态与不完全模态之间的特征一致性, 有效训练缺失模态网络。Yao 等提出了DrFuse [104], 用于利用不完全多模态数据进行有效的医学诊断和预后预测。DrFuse首先学习所有模态之间共享的共性特征, 以及每个独立模态内的模态特定特征, 然后对每个模态的预测结果进行加权并组合, 以生成最终预测。具体而言, 设计了一种按疾病分类的注意力层, 用于生成针对患者和疾病的权重, 以解决模态不一致问题。

一些代表性的不完全多模态学习方法总结于表4。

3.3 讨论

总而言之, 处理不完全数据的方法通常可分为两类: 基于插补的不完全多模态学习和无插补的不完全多模态学习。专注于插补不完全模态的方法可分为两种: 一种通过图或核补全间接恢复缺失模态, 另一种直接填充原始数据。对于无插补的不完全多模态学习方法, 其设计理念更侧重于利用可用模态探索信息和理解多模态数据。然而, 处理不完全多模态数据的方法都面临更深层次的挑战。例如, 通常被忽视的是对缺失模态的插补实例进行质量评估。此外, 使用先验缺失信息来

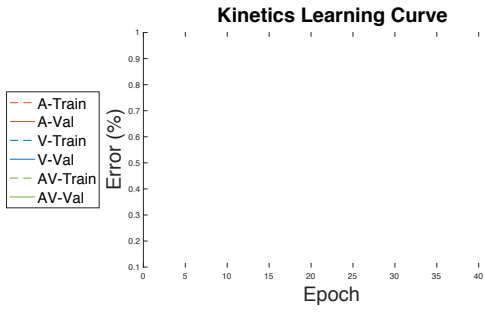


Fig. 3: The learning curves (error-rate) of audio model (A), video model (V), and the naive joint audio-video (AV) model on the Kinetics dataset. Solid lines plot validation error while dashed lines show train error. Figure is from [7]

mask unknown modalities is itself difficult to bridge the information gap and information imbalance issues caused by modal missing. Moreover, incomplete cases such as unpaired views [105], [106], [107] and partially missing labels [108] also need more attention.

4 BALANCED MULTIMODAL LEARNING

Different modalities are tightly correlated, since they depict the same concept from different views. This property inspires the flourish of multimodal learning, where multiple modalities are integrated and expected to enhance the understanding of related events or objects. However, despite the natural cross-modal correlation, each modality has distinct data sources and forms. For instance, audio data is typically represented as one-dimensional waveforms, while visual data consists of images composed of pixels. On the one hand, this discrepancy endows each modality with different properties, such as convergence speed, and then makes it hard to process and learn all modalities well simultaneously, bringing difficulty to joint multimodal learning. On the other hand, this discrepancy also reflects in the quality of unimodal data. Even though all modalities depict the same concepts, their information related to the target event or objects is in different amounts. For example, considering an audio-visual sample of label *meeting*, the visual data apparently show the visual content of the meeting, which is easy to recognize (see Figure.1c). While the corresponding audio data is a noisy street car sound, which is hard to build correlation with the label *meeting*. The information amount of visual modality is clearly more than that of audio modality. Due to the greedy nature of deep neural networks [9], multimodal model would tend to rely on just high-quality modality with sufficient target-related information, while under-fitting the other modalities. To address these challenges and improve the efficacy of multimodal models, recent studies have focused on strategies to balance the differences between modalities and enhance the overall performance of the model.

4.1 Property-discrepancy method

In most multimodal models, different modalities are commonly trained jointly with uniform learning objectives. While this joint training strategy has been generally proven to be effective in practice, there are still potential risk brought

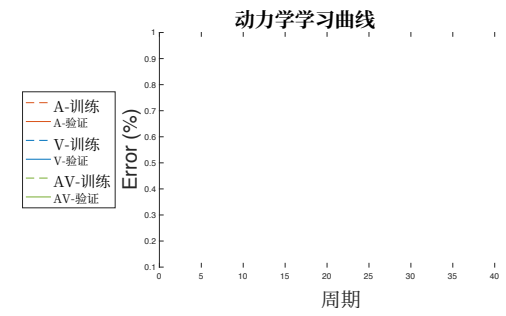


图3: Kinetics数据集上音频模型 (A)、视频模型 (V) 以及朴素联合音频-视频 (AV) 模型的学习曲线 (误差率)。实线绘制验证误差, 虚线显示训练误差。该图来自 [7]

屏蔽未知模态本身难以弥合由模态缺失导致的信息差距和信息不平衡问题。此外, 如成对视图 [105], [106], [107] 和部分缺失标签 [108]等不完整情况也需要更多关注。

4 平衡多模态学习

不同的模态之间具有高度相关性, 因为它们从不同角度描绘了同一个概念。这一特性促进了多模态学习的发展, 其中多种模态被整合起来, 期望能够增强对相关事件或对象的认知。然而, 尽管存在自然的跨模态相关性, 每种模态都有其独特的数据来源和形式。例如, 音频数据通常表示为一维波形, 而视觉数据则由像素组成的图像构成。一方面, 这种差异赋予了每种模态不同的特性, 例如收敛速度, 这使得同时处理和学习所有模态变得困难, 给联合多模态学习带来了挑战。另一方面, 这种差异也体现在单模态数据的质量上。尽管所有模态都描绘了相同的概念, 但它们与目标事件或对象相关的信息量是不同的。例如, 考虑一个标签为<样式 id='1'>会议</样式>的音视频样本, 视觉数据明显展示了会议的视觉内容, 易于识别 (见图1c)。而相应的音频数据是嘈杂的街车声音, 难以与标签<样式 id='3'>会议</样式>建立关联。视觉模态的信息量显然多于音频模态。由于深度神经网络 [9], 多模态模型具有贪婪特性, 往往会依赖仅包含充足目标相关信息的优质模态, 而忽略其他模态, 导致欠拟合。为了应对这些挑战并提高多模态模型的效能, 近期研究主要集中在平衡模态差异的策略上, 以提升模型的整体性能。

4.1 属性差异方法

在大多数多模态模型中, 不同的模态通常使用统一的学习目标进行联合训练。虽然这种联合训练策略在实践中已被普遍证明是有效的, 但仍然存在由多模态数据差异带来的潜在风险。

by the discrepancy in data of multiple modalities. The discrepancy of multimodal data in sources and forms result in that each modality has its own learning property. For example, as shown in Figure 3, in the action recognition dataset Kinetics [110], which contains audio and visual modalities, the audio data tends to require fewer training steps than the vision data before the model begins to be overfitting [111], [112]. The widely-used joint training multimodal framework often neglects this inherent learning property discrepancy of unimodal data, which can potentially lead to negative impacts on the model's performance. More importantly, based on recent study [7], the jointly trained multimodal model even could be outperformed by the best unimodal counterpart. Thus, it is essential to consider the discrepancy in learning properties across each modality to improve the performance of the model. To address this issue and balance the discrepancy in learning properties, several studies have been proposed from different perspectives, including learning-objective-based method, optimization-based method and architecture-based method.

4.1.1 Learning-objective-based method

Since jointly trained multimodal frameworks often only have a uniform loss, which integrates the information of all modalities, the learning of each modality is hard to observe or control. The learning-objective-based methods add unimodal loss and make it possible to targetedly adjust unimodal training:

$$\mathcal{L} = \alpha_{mm}\mathcal{L}_{mm} + \alpha_i \sum_i^n \mathcal{L}_i, \quad (6)$$

where α_{mm} and α_i is the weight of loss function. n is the number of modalities. Since the loss weight determines the influence of the corresponding loss function in parameter update, its selection is essential for controlling unimodal training. Wang *et al.* [7] suggested to determine the optimal loss weights by measuring the proposed overfitting-to-generalization ratio, by which it could comprehensively consider and balance the discrepancy in convergence speed of each modality.

4.1.2 Optimization-based method

Different from the learning-objective-based methods that directly introduce unimodal loss, optimization-based method pays attention to the back-propagation stage. Considering the discrepancy in learning property, optimization-based methods propose to control the unimodal parameter updating by dynamically balancing the learning rates for various modalities [113]:

$$\theta_{t+1}^i = \theta_t^i - \lambda_i g_t^i, \quad (7)$$

where λ_i is the learning rate of the parameter of modality i and g_t^i is the corresponding gradient. In this way, it is possible to conduct modality-specific optimization for each modality. Specifically, Sun *et al.* [113] proposed to balance the optimization of different modalities by assigning a lower learning rate to the modality that is closer to convergence.

4.1.3 Architecture-based method

Besides the above architecture-agnostic methods which focus on learning objective and optimization, methods considering

这种多模态数据在来源和形式上的差异导致每个模态具有其自身的学习特性。例如，如图3所示，在包含音频和视觉模态的动作识别数据集Kinetics [110]，中，在模型开始过拟合 [111], [112]之前，音频数据往往需要的训练步数比视觉数据少。广泛使用的联合训练多模态框架经常忽视单模态数据固有的学习特性差异，这可能会对模型的性能产生负面影响。更重要的是，根据最近的研究 [7]，联合训练的多模态模型甚至可能被最佳的单模态对应模型超越。因此，考虑跨每个模态的学习特性差异对于提高模型性能至关重要。为了解决这个问题并平衡学习特性的差异，从不同角度提出了几种研究方法，包括基于学习目标的、基于优化的和基于架构的方法。

4.1.1 基于学习目标的

由于联合训练的多模态框架通常只有一种统一损失，该损失整合了所有模态的信息，因此难以观察或控制每个模态的学习过程。基于学习目标的方法添加单模态损失，使其能够针对性地调整单模态训练：

$$\mathcal{L} = \alpha_{mm}\mathcal{L}_{mm} + \alpha_i \sum_i^n \mathcal{L}_i, \quad (6)$$

其中 α_{mm} 和 α_i 是损失函数的权重。 n 是模态的数量。由于损失权重决定了相应损失函数在参数更新中的影响，因此其选择对于控制单模态训练至关重要。王 等 [7] 建议通过测量所提出的过拟合到泛化率来确定最佳损失权重，该方法能够全面考虑并平衡各模态收敛速度的差异。

4.1.2 基于优化的

与直接引入单模态损失、基于学习目标的方法不同，基于优化的方法关注反向传播阶段。考虑到学习特性的差异，基于优化的方法提出通过动态平衡各模态的学习率 [113] 来控制单模态参数更新：

$$\theta_{t+1}^i = \theta_t^i - \lambda_i g_t^i, \quad (7)$$

其中 λ_i 是模态 i 参数的学习率， g_t^i 是对应的梯度。这样，就可以对每个模态进行模态特定优化。具体来说，Sun 等人 [113] 提出通过为收敛更近的模态分配较低的学习率来平衡不同模态的优化。

4.1.3 基于架构的方法

除了上述关注学习目标和优化的架构无关方法之外，还有考虑

TABLE 2: A summary of some representativ

Category	Base model
Learning-free imputation	Matrix factorization
	Graph learning
Learning based imputation	Kernel learning
	Raw data imputation
	Matrix factorization
Imputation-free	Projection learning
	Linear regression
	Representation learning
	Graph learning
	Kernel learning
	Matrix factorization
	Weighted fusion
	Representation learning strat
	Weighted fusion
	Representation learning strat

表2：一些代表性不完整多模态学习工作的总

类别	基础模型
无学习插补	矩阵分解
	图学习
基于学习的插补	核学习
	原始数据插补
	矩阵分解
无插补	投影学习
	线性回归
	表征学习
	图学习
	核学习
	矩阵分解
	加权融合
	表征学习策略
	加权融合
	表征学习策略

the discrepancy in learning property problem from the perspective of model architecture are also proposed. At the fusion stage, Xiao *et al.* [112] proposed to randomly drop the network pathway of the modality with faster convergence speed altogether with a certain probability. In this way, the training of the corresponding modality is slowed down and its learning dynamics are ensured to be more compatible with other unimodal counterparts. Besides the late fusion architecture, studies that utilize low-level cross-modal fusion to enhance multimodal joint learning are also proposed. Zhou *et al.* [114] utilized central connections to link low-level features in both video and audio streams, capturing spatial (for video frames) and temporal information, thereby achieving high-level semantic representations and improving joint multimodal learning.

4.2 Quality-discrepancy methods

In addition to the issue of discrepancy in unimodal learning property, even though all modalities convey the same concepts, they contain varying amounts of information related to the target event or objects, resulting a discrepancy in unimodal data quality. Given the inherently greedy nature of deep neural networks, recent studies have further uncovered that multimodal models tend to rely heavily on a single

除了上述后期融合架构之外，还有利用低层跨模态融合来增强多模态联合学习的研究也被提出。Zhou 等 [114]利用中心连接将视频和音频流中的低层特征连接起来，捕捉空间（针对视频帧）和时间信息，从而实现高级语义表示并提升联合多模态学习。

4.2 质量差异方法

除了单模态学习特性中的差异问题外，即使所有模态都传达相同的概念，它们包含与目标事件或对象相关的信息量也不同，导致单模态数据质量存在差异。鉴于深度神经网络的天性具有贪婪性，最近的研究进一步发现，多模态模型往往过度依赖单一

tive incomplete multimodal learning works.

	Task	Reference
n	Clustering	[43], [45]
		[46], [47], [48]
		[57]
		[50]
		[38], [52]
		[51] [53], [55], [56]
	Modality generation	[58]
		[59]
		[60]
		[62], [63], [64]
ng	Breast cancer prediction	[61]
	Representation learning	[71]
	Action recognition	[109]
		[66], [67]
	Clustering	[68]
		[75], [44], [76]
		[78], [79]
	Regression	[82]
		[85]
	Classification	[83]
ategy	Alzheimer’s disease prediction	[89]
		[77]
		[87]
	Clustering	[86], [88]
	Alzheimer’s disease diagnosis	[90]
		[91], [92], [93]
	Clustering	[94]
		[96], [97], [99]
	Classification	[100], [102], [104]
		[101]
ategy	Medical image segmentation	[103]

总结。

Task	参考文献
聚类	[43], [45]
	[46], [47], [48]
	[57]
	[50]
	[38], [52]
	[51] [53], [55], [56]
模态生成	[58]
	[59]
	[60]
	[62], [63], [64]
乳腺癌预测	[61]
表征学习	[71]
动作识别	[109]
	[66], [67]
聚类	[68]
	[75], [44], [76]
	[78], [79]
回归	[82]
	[85]
分类	[83]
阿尔茨海默病预测	[89]
	[77]
	[87]
聚类	[86], [88]
阿尔茨海默病诊断	[90]
	[91], [92], [93]
	[94]
	[96], [97], [99]
分类	[100], [102], [104]
	[101]
医学图像分割	[103]

modality with high quality that provides sufficient target-related information, while neglecting the others, leading to their under-optimized [8], [10]. As shown in Fig. 4, Peng *et al.* [8], [128] empirically verified that the quality of unimodal representations in the multimodal model is worse than that in the corresponding unimodal models, despite the overall superior performance of the multimodal models. Moreover, this under-optimization of each modality is imbalanced, and the modality with high quality is less under-optimized. They claim that this imbalanced problem is caused by the model preference for the high-quality modality with better performance (i.e., the dominant modality). For instance, the audio modality in the curated sound-oriented dataset, VGGSound [129], is less under-optimized compared to the vision modality. Theoretical analysis conducted by Huang *et al.* [10] has further confirmed that for the jointly trained multimodal late-fusion framework, the encoder networks can only learn a subset of modalities, and other modalities may fail to be discovered well. This finding is consistent with empirical observations, indicating that the imbalanced preference for different modalities issue needs to be addressed in multimodal learning. To tackle this problem brought by discrepancies in unimodal data quality, a series of studies have been proposed, encompassing learning-objective-based method, optimization-based

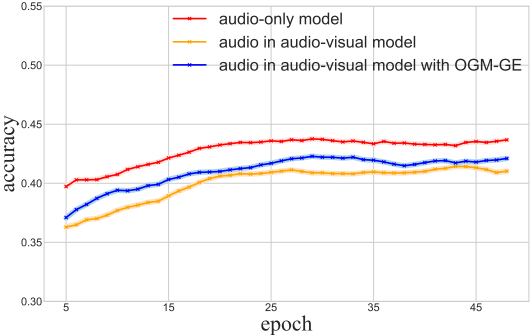
一种高质量模态，能够提供充足的与目标相关的信息，而忽略其他模态，导致其他模态的欠优化 [8], [10]。如图4所示，Peng 等人 [8], [128] 通过实验验证了，尽管多模态模型的整体性能更优，但其单模态表示的质量仍劣于相应的单模态模型。此外，各模态的欠优化程度并不均衡，高质量模态的欠优化程度相对较轻。他们认为，这种不均衡问题是由模型对性能更优（即主导模态）的高质量模态的偏好所导致的。例如，在精选的以声音为导向的数据集VGGSound [129]中，音频模态的欠优化程度低于视觉模态。Huang 等人 [10]进行的理论分析进一步证实，对于联合训练的多模态晚期融合框架，编码器网络只能学习部分模态，而其他模态可能无法得到充分挖掘。这一发现与经验观察结果一致，表明在多模态学习中，需要解决对不同模态的不均衡偏好问题。为了应对单模态数据质量差异带来的问题，一系列研究已被提出，包括基于学习目标的、基于优化的、基于架构的和基于数据增强的方法。

TABLE 3: A summary of representative b

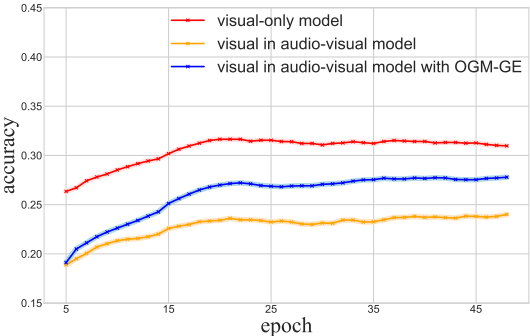
表3：代表性平衡多模态学习研究的总结

Category		
Property discrepancy	Optimization-based	
	Architecture-based	
Quality discrepancy	Learning-objective-based	Audi
	Optimization-based	Actio
	Architecture-based	Multi R
	Data-augmentation-based	Muli-vie Action

类别		
属性差异	基于优化的	
	基于架构的	
质量差异	基于学习目标的	视听
	基于优化的	动作
	基于架构的	多模态 R
	基于数据增强的	多视角分 动作分

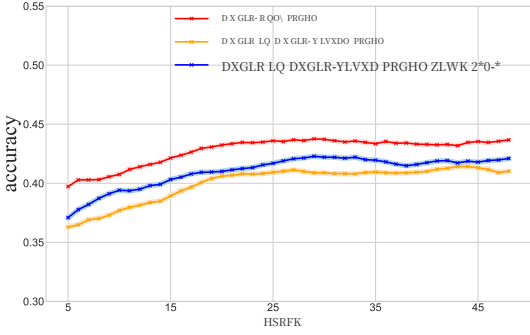


(a) Audio.

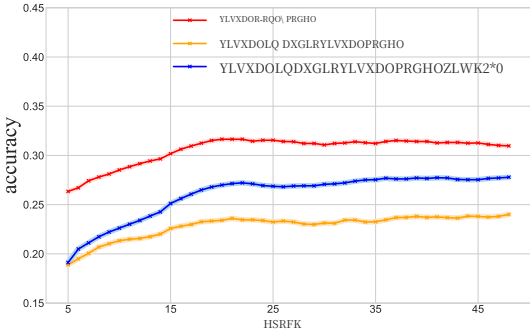


(b) Visual.

Fig. 4: Performance of the uni-modal models, joint-trained multimodal model, and multimodal model with OGM-GE [8] on the validation set of the VGGSound dataset. Figures are from [8].



(a) 音频。



(b) 视觉。

图4：单模态模型、联合训练的多模态模型以及带有 OGM-GE [8]的多模态模型在VGGSound数据集验证集上的性能。图示来自 [8]。

method, architecture-based method and data-augmentation-based method.

方法、基于架构的方法和基于数据增强的方法。

e balanced multimodal learning studies.

结。

Application	Reference	应用	参考文献
Action classification	[7]	动作分类	[7]
Sentiment analysis	[113]	情感分析	[113]
Action classification	[112]	动作分类	[112]
Deepfake detection	[114]	Deepfake检测	[114]
Violence Detection	[115]	暴力检测	[115]
Action classification	[116]	动作分类	[116]
Product understanding	[117]	产品理解	[117]
Audio-visual fine-grained classification	[118]	听细粒度分类	[118]
Action classification	[119]	动作分类	[119]
Audio-visual fine-grained classification, sentiment analysis	[8], [120]	动作分类, 情感分析	[8], [120]
Rumor detection	[121]	谣言检测	[121]
Audio-visual video parsing	[122]	视听视频解析	[122]
Sentiment analysis	[123]	情感分析	[123]
Audio-visual Conditional Image Synthesis	[124]	条件图像合成	[124]
RGB-T Semantic Segmentation	[125]	RGB-T语义分割	[125]
Audio-visual fine-grained classification, gesture classification	[9]	分类, 手势分类	[9]
Emotion recognition	[126]	情感识别	[126]
Audio-visual fine-grained classification, emotion recognition	[127]	分类, 情感识别	[127]

4.2.1 Learning-objective-based method

Considering the preference of multimodal model for modality with high quality, several methods which introduce additional learning objectives beyond normal joint learning loss function are proposed. These methods tend to utilize additional loss to break the modality preference of multimodal model:

$$\mathcal{L} = \mathcal{L}_{mm} + \mathcal{L}_{uni}, \quad (8)$$

where $\mathcal{L}_{mm}$ is the multimodal loss function and $\mathcal{L}_{uni}$ is a specifically designed loss function that improves unimodal learning, especially the learning of modality with lower quality. Concretely, Yang *et al.* [115] endowed the multimodal contrastive learning loss to constrain the audio and visual modality to be as close as possible in the semantic space, then help to make full use of all modalities. Also paying attention to the multimodal semantic space, Ma *et al.* [130] proposed a novel regularization technique Calibrating Multimodal Learning, to enforce the consistency between prediction confidence and the number of modalities, avoiding the issue that multimodal classification models tend to over-rely on specific modalities. In addition to these methods in semantic space, some methods introduce the learning objective of knowledge distillation. Du *et al.* [116] proposed to distill the pre-trained uni-modal features to the corresponding parts of multimodal late-fusion models, to avoid the learning preference for one modality. Differently, Liu *et al.* [117] designed a self-distillation training strategy, which can automatically teach less-optimized modality from better-optimized modality. Besides these common multimodal classification task, Xu *et al.* [118] focused on the audio-visual fine-grained classification task, which has a higher demand for distinguishable feature distribution. They proposed the multimodal Cosine loss, MMCosine, which performs a modality-wise $L2$ normalization to features and weights towards balanced and better multimodal fine-grained learning. Beyond the generalization performance, Yang *et al.* [119] put attention on the multi-modal robustness. They emphasized that multi-

4.2.1 基于学习目标的

考虑到多模态模型对高质量模态的偏好, 研究者提出了几种在正常联合学习损失函数之外引入额外学习目标的方法。这些方法倾向于利用额外损失来打破多模态模型的模态偏好:

$$\mathcal{L} = \mathcal{L}_{mm} + \mathcal{L}_{uni}, \quad (8)$$

在 $\mathcal{L}_{mm}$ 是多模态损失函数, 而 $\mathcal{L}_{uni}$ 是一种专门设计的损失函数, 可以改进单模态学习, 特别是对质量较低模态的学习。具体来说, Yang等 [115] 将多模态对比学习损失用于约束音频和视觉模态在语义空间中尽可能接近, 从而帮助充分利用所有模态。同时关注多模态语义空间, Ma 等. [130]提出了一种新的正则化技术 Calibrating Multimodal Learning, 以强制预测置信度与模态数量之间的一致性, 避免多模态分类模型倾向于过度依赖特定模态的问题。除了语义空间中的这些方法, 一些方法引入了知识蒸馏的学习目标。Du 等 [116]提出将预训练的单模态特征蒸馏到多模态晚期融合模型的相应部分, 以避免对单一模态的学习偏好。不同地, Liu 等 [117] 设计了一种自蒸馏训练策略, 可以从优化更好的模态自动教导优化较差的模态。除了这些常见的多模态分类任务, Xu等 [118] 关注于音频-视觉细粒度分类任务, 该任务对可区分特征分布有更高的需求。他们提出了多模态余弦损失, MMCosine, 它对特征和权重进行模态 $L2$ 归一化, 以实现平衡和更好的多模态细粒度学习。除了泛化性能, Yang 等 [119] 关注于多模态鲁棒性。他们强调多模态

modal models are often vulnerable to attacks on the specific modality, and introduced margin-based regularization to ensure multi-modal robustness.

4.2.2 Optimization-based method

Optimization-based methods focus on the back-propagation stage. These methods respectively pay attention to the magnitude and direction of unimodal gradient, to improve the learning of modality with lower quality. To slow down the learning of preferred modality with high quality and accordingly offer more training efforts to other modalities, Peng and Wei *et al.* [8], [128] proposed an on-the-fly gradient modulation strategy, which dynamically monitors the contribution difference of various modalities to the final prediction during training and mitigates the gradient magnitude of dominant modality to focus more on other modalities. Inspired by this gradient magnitude modulation idea, Sun *et al.* [121] extended it into the graph-based models of rumor detection task, adaptively adjusting the magnitude of gradients to alleviate the preference for high-quality modality. Similarly, Fu *et al.* [122] extended the idea of gradient magnitude modulation into audio-visual video parsing task. They further proposed a modality-separated decision unit to well evaluate the unimodal prediction in the more complex audio-visual video parsing framework with more cross-modal interaction. Besides the gradient magnitude, the direction of unimodal gradient is also explored. Fan *et al.* [120] suggested that the slow-learning modality's updating direction is severely disturbed by the preferred high-quality modality, making it hard to its training. Subsequently, they utilize prototypes, which are the centroids of each class within the representation space, to refine the updating direction, aiming to improve unimodal performance.

4.2.3 Architecture-based method

Architecture-based methods aim to balance the training of modalities with discrepancy in quality by designing better unimodal representation learning modules. He *et al.* [123] designed a Multimodal Temporal Attention module, which takes into account the temporal influences of all modalities on each individual unimodal branch. This module facilitates the interaction among unimodal branches and achieves adaptive inter-modal balance. Su *et al.* [124] focused on the generative scenarios, and utilized the coordinated feature space, named Coordinated Knowledge Mining, to improve the under-optimized modalities with the guidance of preferred high-quality modalities. Lin *et al.* [125] proposed a novel Variational Feature Fusion Module, which regards fusion features as random variables and obtains more balanced segmentation performance for different categories and modalities.

4.2.4 Data-augmentation-based method

Data-augmentation-based methods are with the idea of enhancing low-quality modality at the data input stage. The basic intuition is that solely training of unimodal data could avoid the influence of other modalities. There are two kinds of strategies, introducing additional individual training for less learnt low-quality modality or removing the data of better learnt high-quality modality. For the former idea, Wu *et*

模态模型通常容易受到针对特定模态的攻击，并引入了基于边界的正则化方法来确保多模态鲁棒性。

4.2.2 基于优化的方法

基于优化的方法聚焦于反向传播阶段。这些方法分别关注单模态梯度的幅值和方向，以提升低质量模态的学习效果。为减缓高质量模态的学习速度，并相应地为其他模态提供更多训练资源，彭和魏等 [8], [128] 提出了一种动态梯度调制策略，该策略在训练过程中动态监控各种模态对最终预测的贡献差异，并降低主导模态的梯度幅值，以更关注其他模态。受此梯度幅值调制思路的启发，孙等 [121] 将其扩展到谣言检测任务的基于图的模型中，自适应地调整梯度幅值，以减轻对高质量模态的偏好。类似地，付等 [122] 将梯度幅值调制的思想扩展到视听视频解析任务中。他们进一步提出了一种模态分离决策单元，以在具有更多跨模态交互的更复杂的视听视频解析框架中更好地评估单模态预测。除了梯度幅值，单模态梯度的方向也得到了探索。范等 [120] 指出，学习较慢模态的更新方向受到偏好高质量模态的严重干扰，使其难以训练。随后，他们利用原型（表示空间中每个类别的中心点）来细化更新方向，旨在提升单模态性能。

4.2.3 基于架构的方法

基于架构的方法旨在通过设计更好的单模态表征学习模块来平衡不同模态训练时的质量差异。He 等 [123] 设计了一个多模态时序注意力模块，该模块考虑了所有模态对每个单模态分支的时序影响。该模块促进了单模态分支之间的交互，并实现了自适应的模态间平衡。Su 等 [124] 专注于生成场景，并利用了协调特征空间（名为协调知识挖掘），在高质量模态的指导下改进了欠优化的模态。Lin 等 [125] 提出了一种新颖的变分特征融合模块，该模块将融合特征视为随机变量，并为不同类别和模态获得了更平衡的分割性能。

4.2.4 基于数据增强的方法

基于数据增强的方法在数据输入阶段旨在增强低质量模态。其基本原理是，仅训练单模态数据可以避免其他模态的影响。存在两种策略，为学习较少的低质量模态引入额外的单独训练，或移除学习较好的高质量模态的数据。对于前者，Wu *et*

al. [9] firstly measured the relative speed at which the model learns from one modality compared to other modalities, and then introduced the re-balancing steps at which update the underutilized uni-modal branches intentionally in order to accelerate the model to learn from its input modality. For the latter one, Zhou *et al.* [126] suggested adaptively masking the data input of well-learned modality, accordingly encouraging the model to fit other modalities better. Most imbalanced multi-modal learning methods ignore the fine-grained modality discrepancy, where the dominant modality could vary across different samples under realistic scenarios. Wei *et al.* [127] introduced a Shapley-based sample-level modality valuation metric, to observe the contribution of each modality during prediction for each sample, and then dynamically re-sample low-contributing modalities during training. This method achieves fine-grained multi-modal cooperation.

4.3 Discussion

Overall, balanced multimodal learning methods focus on the discrepancy in learning property or data quality among different modalities, which is caused by the heterogeneity of multimodal data. These methods propose solutions from different perspectives, including learning objective, optimization, architecture, and data augmentation. The balanced multimodal learning is still a sunrise field, with plenty of under-explored directions. For example, it is promising to explore the cooperation between modalities with theoretical guidance. Moreover, the current methods mainly limit to the typical multimodal task, mostly discriminative tasks and a few generative tasks. In fact, beyond these, the Multimodal Large Language Model, which also needs to jointly integrate different modalities, could also suffer from this imbalance problem. It is expected to extend current studies or design new solutions in Multimodal Large Language Model settings.

5 DYNAMIC MULTIMODAL FUSION

Due to unexpected environmental factors and sensor issues, some modalities may suffer from poor reliability and lack of task-specific information. Besides, the quality of different modalities are dynamically varying from scenarios. We borrow a case from [131], as shown in Fig. 5, thermal modality tends to provide more task-specific information compared to RGB modality during nighttime, whereas the situation is reversed during the daytime. This phenomenon motivates a new multimodal learning paradigm, i.e., dynamical multimodal fusion, which aims to adapt to the dynamically changing quality of multimodal data and selectively integrate task-specific information. In this section, we focus on the challenges of dynamical multimodal fusion and categorize current advances in this literature into three main lines including heuristic, attention-based and uncertainty-aware dynamic fusion.

5.1 Heuristic dynamic fusion

Toward achieving robust multimodal fusion, previous works heuristically devise dynamic fusion strategies. These dynamic fusion methods are derived from human experience

al. [9] 首先测量了模型从一种模态学习相对于其他模态的相对速度, 然后引入了重新平衡的步骤, 在步骤中故意更新未充分利用的单一模态分支, 以加速模型从其输入模态学习。对于后者, Zhou 等人。[126]建议自适应地屏蔽已学习良好的模态的数据输入, 相应地鼓励模型更好地拟合其他模态。大多数不平衡的多模态学习方法忽略了细粒度的模态差异, 在非现实场景下, 主导模态可能在不同样本中变化。Wei等人。[127] 引入了一种基于Shapley的样本级模态评估指标, 以观察每个样本在预测过程中每个模态的贡献, 然后在训练过程中动态重新采样低贡献模态。这种方法实现了细粒度的多模态协作。

4.3 讨论

总体而言, 平衡多模态学习方法关注不同模态在学习特性或数据质量上的差异, 这是由多模态数据的异构性引起的。这些方法从不同角度提出了解决方案, 包括学习目标、优化、架构和数据增强。平衡多模态学习仍然是一个充满希望的新兴领域, 有许多尚未充分探索的方向。例如, 在理论指导下探索模态间的合作具有前景。此外, 当前方法主要局限于典型的多模态任务, 大多是判别性任务和少量生成性任务。实际上, 除了这些之外, 也需要联合集成不同模态的多模态大语言模型也可能受到这种不平衡问题的影响。预计需要在多模态大语言模型设置中扩展当前研究或设计新的解决方案。

5 动态多模态融合

由于意外的环境因素和传感器问题, 某些模态可能存在可靠性差和缺乏任务特定信息的问题。此外, 不同模态的质量会随着场景动态变化。我们借鉴 [131], 中的案例 (如图5所示) 在夜间, 热成像模态往往比RGB模态提供更多任务特定信息, 而在白天则相反。这种现象催生了一种新的多模态学习范式, 即动态多模态融合, 旨在适应多模态数据的动态变化质量, 并选择性地整合任务特定信息。在本节中, 我们关注动态多模态融合的挑战, 并将该领域当前的研究进展分为三大主线: 启发式、基于注意力的和不确定性感知的动态融合。

5.1 启发式动态融合

为实现鲁棒的多模态融合, 以往工作启发式地设计动态融合策略。这些动态融合方法源于人类经验

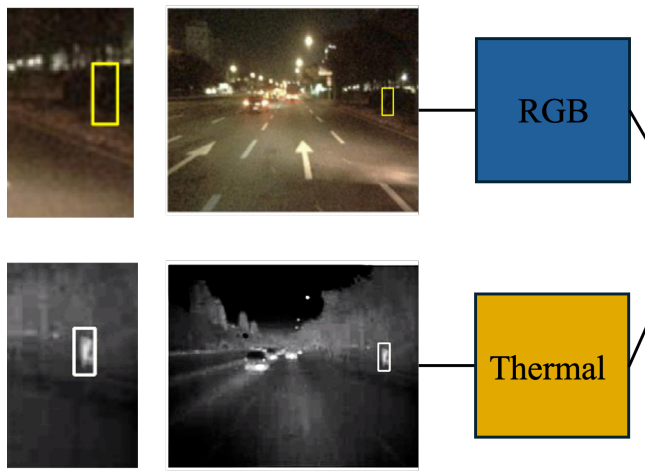


Fig. 5: Illustrations o

and knowledge (e.g., thermal modality is more reliable than RGB in nighttime) about the task at hand and application scenarios. As a concrete example, illumination conditions can serve as a criterion to introduce dynamic fusion. In around-the-clock applications, previous work identifies that the quality of RGB and thermal modalities are temporally varying in multispectral pedestrian detection task [131]. While the RGB modality tends to contain more useful information compared to the thermal modality under normal illumination conditions, this relationship can be reversed under low light or nighttime conditions where the thermal modality becomes more reliable than RGB. Based on this observation, the authors proposed to utilize an illumination aware fusion module that fuse features from the two modalities adaptively. Similarly, to enable adaptation to varying illumination conditions, Guan *et al.* [132] proposed an illumination-aware weighting mechanism that evaluates the lighting of each input scene. These illumination weights are utilized to dynamically integrate the outputs of sub-networks specialized for day and night detection and segmentation. The proposed method can further be applied to pedestrian detection in security surveillance and autonomous driving systems.

Besides environmental factors like illumination, the criterion of dynamic fusion can be devised according to knowledge about the properties of neural networks. When vision models are involved, activation level of the feature maps and the scaling factors in batch-norm layers can indicate input informativeness. Inspired by this, Li *et al.* [133] proposed an encoder-decoder style network with dynamic fusion strategy. In their implementation, the l1-norm of feature maps is calculated as channel-wise importance, which is then normalized to form a weighting vector that selectively emphasizes informative features from each modality. Wang *et al.* [111] proposed Channel Exchanging Networks (CEN) that dynamically exchange information between modality sub-networks based on channel importance evaluated by the scaling factors of batch-normalize layers. By performing channel exchange within modality-specific regions and sharing convolutional filters, CEN balances unimodal feature learning and multimodal fusion dynamically.

Some relevant works also utilize task-specific modules to

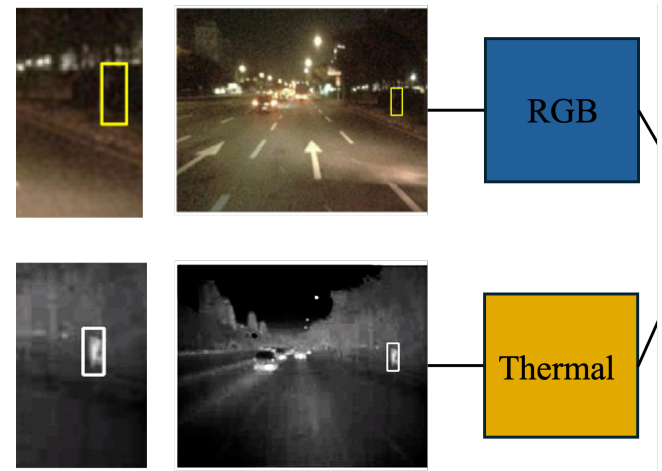
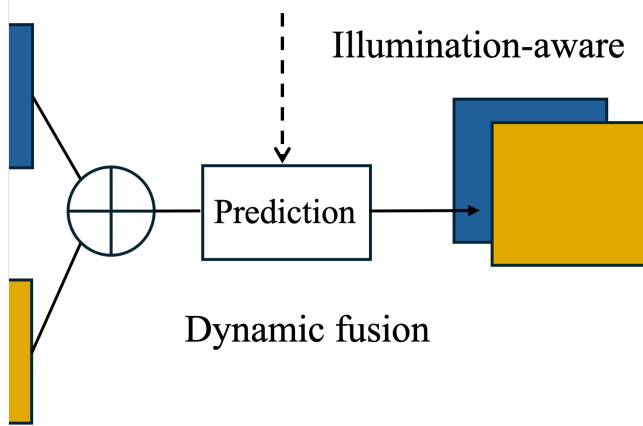


图5: 动态融合的示

为实现鲁棒的多模态融合，以往研究通过启发式方法设计动态融合策略。这些动态融合方法源于对任务本身和应用场景的人体经验和知识（例如，夜间热成像模态比RGB模态更可靠），并据此进行设计。以具体实例为例，光照条件可作为引入动态融合的标准。在全天候应用中，以往研究指出，在多光谱行人检测任务中，RGB和热成像模态的质量会随时间变化。在正常光照条件下，RGB模态通常比热成像模态包含更多信息，但在低光照或夜间条件下，热成像模态的可靠性会超过RGB。基于这一观察，作者提出了一种光照感知融合模块，能够自适应地融合两种模态的特征。类似地，为适应变化的光照条件，Guan 等。 [132] 提出了一种光照感知加权机制，用于评估每个输入场景的光照情况。这些光照权重被用于动态整合专门用于日间和夜间检测与分割的子网络的输出。该方法还可进一步应用于安防监控和自动驾驶系统中的行人检测。

除了光照等环境因素，动态融合的标准可以根据关于神经网络特性的知识来制定。当涉及视觉模型时，特征图的激活级别和批量归一化层中的缩放因子可以指示输入的信息量。受此启发，Li 等 [133]提出了一种带有动态融合策略的编码器-解码器风格网络。在他们的实现中，特征图的l1范数被计算为通道级重要性，然后被归一化以形成一个权重向量，该向量会选择性地强调来自每个模态的 informative 特征。Wang 等 [111] 提出了通道交换网络（CEN），该网络根据批量归一化层的缩放因子评估的通道重要性，在模态子网络之间动态交换信息。通过在特定模态区域内进行通道交换并共享卷积滤波器，CEN 动态地平衡了单模态特征学习和多模态融合。

一些相关研究也利用任务特定模块来



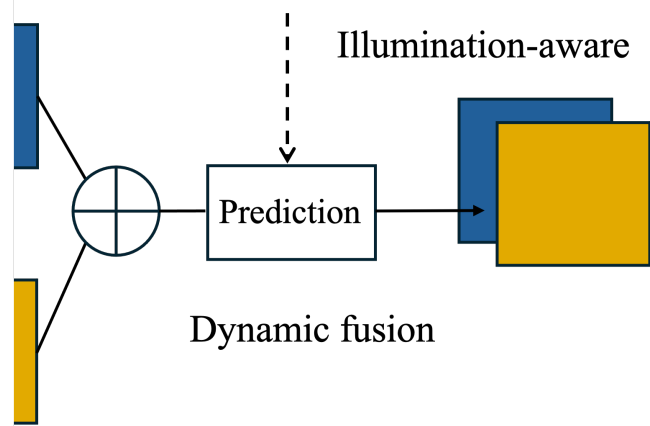
s of dynamic fusion.

achieve dynamic fusion. To highlight a few, Panda *et al.* [134] introduced AdaMML, an adaptive multimodal learning in efficient video recognition. By dynamically selecting optimal input modalities per video segment, this work contributes to data-dependent multimodal selection for efficient video recognition. In this work, a lightweight policy network is devised to dynamically select optimal input modalities for each video segment. AdaMML utilizes Gumbel-Softmax sampling for differentiable training of discrete decisions. Similarly, Xue *et al.* [136] proposed DynMM, a novel approach for dynamic multimodal fusion. It introduces modality-level and fusion-level decisions, allowing adaptive selection of input modalities and fusion operations based on data characteristics.

5.2 Attention-based dynamic fusion

The key challenge of dynamic fusion is to design dynamic mechanism to learn a reasonable fusion criterion. To achieve this, another line of methods typically introduce attention mechanism to dynamic fusing multimodal information. In this subsection, we organize various dynamic fusion methods according to different types of involved attention mechanism, including self-attention, channel-attention, spatial-attention and transformer.

Self-attention models dependency within the input sequence by allowing each element to interact with others. In multimodal emotion recognition (involving textual and audio modalities), audio modality tend to convey more task-specific information than text, such as intonation. However, due to the potential influence of background noise, the quality of these two modalities may vary significantly from different samples. Sun *et al.* introduced a novel Multimodal Cross and Self-Attention Network (MCSAN) [137] to dynamically emphasize information from both the linguistic content and acoustic information in speech emotion recognition. Similarly, Hazarika *et al.* devised a self-attention based multimodal fusion strategy to adapt to the varying quality of textual and audio modalities in emotion recognition task [156]. By weighing modalities at the feature level, the proposed method provides extra stability against noisy inputs.



意图。

实现动态融合。举几例来说，Panda 等人。[134]提出了 AdaMML，这是一种用于高效视频识别的自适应多模态学习方法。通过动态选择每个视频片段的最佳输入模态，这项工作作为高效视频识别的数据相关多模态选择做出了贡献。在这项工作中，设计了一种轻量级的策略网络，用于动态选择每个视频片段的最佳输入模态。AdaMML使用 Gumbel-Softmax采样进行离散决策的可微分训练。类似地，Xue 等人。[136]提出了DynMM，这是一种新颖的动态多模态融合方法。它引入了模态级别和融合级别的决策，允许根据数据特征自适应选择输入模态和融合操作。

5.2 基于注意力的动态融合

动态融合的关键挑战在于设计动态机制来学习合理的融合标准。为实现此目标，另一类方法通常引入注意力机制来动态融合多模态信息。在本小节中，我们根据所涉及的注意力机制类型，对各种动态融合方法进行组织，包括自注意力、通道注意力、空间注意力和Transformer。

自注意力模型通过允许每个元素与其他元素交互，依赖输入序列中的模型。在多模态情感识别（涉及文本和音频模态）中，音频模态往往比文本传递更多特定任务的信息，例如语调。然而，由于背景噪声的潜在影响，这两种模态的质量在不同样本中可能会有显著差异。Sun 等引入了一种新的多模态交叉自注意力网络（MCSAN）[137]，用于动态强调语音情感识别中语言内容和声学信息的信息。类似地，Hazarika 等设计了一种基于自注意力的多模态融合策略，以适应情感识别任务中文本和音频模态质量的变化 [156]。通过在特征层权衡模态，所提出的方法提供了额外的稳定性，以抵抗噪声输入。

TABLE 4: A summary of our taxono

表4: 我们动态多模态融合分类法的总

Type	Criterion		Type	标准	
Heuristic	Illumination condition	A	启发式方法	光照条件	
	Illumination condition			光照条件	
	Reinforcement learning			强化学习	
	Activity map			活动图	
	Correlation			相关性	
	Policy Network			策略网络	
Attention-based	Feature confidence	Au	基于注意力的	特征置信度	视
	Gating Networks			门控网络	
	Self-attention			自注意力	
	Spatial attention			空间注意力	
	Channel attention			通道注意力	
Uncertainty-aware	Transformer			Transformer	
	Entropy		不确定性感知	熵	
	Energy score			能量分数	
	Bayes’ rule			贝叶斯法则	
	Subjective Logic	Mu		主观逻辑	多
	Gaussian distribution			高斯分布	
	NIG Distributions			NIG 分布	
	Prediction confidence			预测置信度	
	Gaussian Process			高斯过程	

Usually applied in vision tasks, channel attention dynamically evaluates the importance of different channels to achieve better performance. To adaptively emphasize important multimodal features while suppressing less relevant ones, Hamid *et al.* [139] introduced a Multimodal Transfer Module (MMTM) to achieve better overall performance. MMTM is designed for intermediate fusion and can be inserted at different levels of the feature hierarchy, allowing recalibration of channel-wise features of low-quality modality with knowledge from others. A squeeze-then-excitation operation is implemented to generate a joint representation and excitation weights, enabling adaptive feature recalibration.

The motivation of spatial attention is to assess the importance of different locality in the feature map from the channel dimension. Cao *et al.* introduced a novel lightweight fusion module called Channel Switching and Spatial Attention (CSSA) for multimodal object detection [138]. The proposed CSSA module efficiently fuses inputs from different

通常应用于视觉任务，通道注意力动态评估不同通道的重要性，以实现更好的性能。为了自适应地强调重要的多模态特征，同时抑制不太相关的特征，Hamid 等 [139] 引入了一种多模态转换模块（MMTM）以实现更好的整体性能。MMTM 设计用于中间融合，可以插入特征层次结构的不同级别，允许用来自其他模态的知识重新校准低质量模态的通道特征。实现了一种挤压-激励操作，以生成联合表示和激励权重，从而实现自适应特征重新校准。

空间注意力的动机是从通道维度评估特征图中不同局部区域的重要性。曹等。为多模态目标检测 [138]引入了一种新颖的轻量级融合模块，称为通道切换和空间注意力（CSSA）。所提出的CSSA模块有效地融合了来自不同

omy of dynamic multimodal fusion .

Application	Reference
Pedestrian detection	[132]
Pedestrian detection	[131]
Information extraction	[11]
Image fusion	[133]
Semantic Segmentation	[111]
Image-to-image translation	[134]
Audio-visual action recognition	[135]
Multi-omics classification	[136]
Sentiment analysis	[137]
Semantic segmentation	[138]
Movie genre classification	[139]
Emotion recognition	[137]
Object detection	[138]
Audio-visual speech enhancement	[139]
Human action recognition	[139]
Human gesture recognition	[140]
Semantic segmentation	[140]
Classification and retrieval	[141]
Image-to-image translation	[141]
RGBD semantic segmentation	[142]
3D object detection	[142]
Classification	[143]
Semantic segmentation	[144]
Sense recognition	[145]
Object detection	[146]
Sense recognition	[146]
Image-text classification	[2], [147], [148]
Object detection	[149], [150]
Medical image classification	[149], [150]
Semantic segmentation	[151], [152]
Emotion recognition	[151], [152]
Sentiment recognition	[153]
Multi-omics classification	[26], [154]
Multi-spectral pedestrian detection	[26], [154]
Multi-view classification	[155]

- modalities using spatial attention. In their implementation, spatial attention is enhanced using max and average pooling
- to retain unique features while effectively fusing information.

As one of the most advanced variants of self-attention, transformers (based on multi-head self-attention) are frequently used in recent multimodal fusion methods to enhance the expressive power of the model. Nagrani *et al.* proposed Fusion Bottlenecks, a new transformer architecture that restricts cross-modal interactions to bottleneck tokens and mid-late fusion layers only [140]. Fusion Bottlenecks force the model to share only necessary information between modalities, avoiding expensive full pairwise attention. Key to the approach is the insight that full pairwise attention between all modalities’ tokens is often redundant, and condensing interactions through fusion bottlenecks provides an efficient alternative. Girdhar *et al.* leveraged a shared Transformer architecture with shared parameters to handle diverse vision modalities including images, videos, and single-view 3D data.

总结。

应用	参考文献
行人检测	[132]
行人检测	[131]
信息提取	[11]
图像融合	[133]
语义分割	[111]
图像到图像翻译	[134]
视听动作识别	[134]
多组学分类	[135]
情感分析	[136]
语义分割	[136]
电影类型分类	[136]
情感识别	[137]
目标检测	[138]
视听语音增强	[139]
人类动作识别	[139]
人类手势识别	[140]
语义分割	[140]
分类与检索	[141]
图像到图像翻译	[141]
RGBD语义分割	[142]
3D目标检测	[142]
分类	[143]
语义分割	[144]
感知识别	[145]
目标检测	[146]
感知识别	[146]
图像文本分类	[2], [147], [148]
目标检测	[149], [150]
医学图像分类	[149], [150]
语义分割	[151], [152]
情感识别	[151], [152]
情感识别	[153]
多组学分类	[26], [154]
多光谱行人检测	[26], [154]
多视角分类	[155]

使用空间注意力融合模态。在他们的实现中，使用最大池化和平均池化增强空间注意力，以保留独特特征的同时有效融合信息。

作为自注意力最先进的变体之一，Transformer（基于多头自注意力）经常被用于近期的多模态融合方法中，以增强模型的表达能力。Nagrani 等人提出了Fusion Bottlenecks，这是一种新的Transformer架构，它将跨模态交互限制在瓶颈token和中期后期融合层中 [140]。Fusion Bottlenecks迫使模型只在模态之间共享必要的信息，避免了昂贵的全对全注意力。该方法的关键在于，所有模态的token之间的全对全注意力往往是冗余的，而通过融合瓶颈压缩交互提供了一种高效的替代方案。Girdhar 等人利用了一个共享参数的Transformer架构来处理包括图像、视频和单视图3D数据在内的各种视觉模态。

The key idea is to fuse information from different modalities into a shared embedding representation by employing a self-attention mechanism to model cross-modality relationship both temporally and spatially [141]. Wang *et al.* identified that existing Transformer variants might dilute unimodal modal attention weights on multimodal tasks, leading to sub-optimal overall performance. To overcome this, they propose to dynamically identifies uninformative tokens and replace them with aggregated features from other modalities for effective fusion [142]. Unlike prior works that mainly focus on enhancing reliability of multimodal fusion during training time, Yang *et al.* [157] proposed to utilizing attention to fully handle corrupted modalities in test time. A confidence-aware loss is devised to enhance credible predictions and mitigate potential influence of noise.

5.3 Uncertainty-aware dynamic fusion

Contrast to heuristic dynamic multimodal learning methods based on intuitive assumptions which might not always hold in practice, uncertainty-based multimodal fusion emerges as a more general and principled way to achieve reliable fusion very recently, which is usually built on solid foundations, e.g., probability distribution or information theory.

Subjective Logic is a commonly used uncertainty estimation method in classification task and a framework that relates the parameters of a Dirichlet distribution to a belief distribution. The Dirichlet distribution can be considered as the conjugate prior distribution of a categorical distribution. SL provides a powerful tool for reasoning under uncertainty and making decisions based on incomplete or uncertain information. Motivated by SL, a Trustworthy Multimodal Classification (TMC) [147] method was proposed recently based on subjective logic and evidence theory. It provides a new paradigm for multimodal classification by dynamically integrating different modalities with Dempster-Shafer (D-S) theory. Taking two modalities scenario as an example, the multimodal belief masses are calculated in the following manner:

$$\mathcal{M} = \mathcal{M}^1 \oplus \mathcal{M}^2, \quad (9)$$

where $\mathcal{M} = \{\{b_k\}_{k=1}^K, u\}$ is the multimodal brief mass, $\mathcal{M}^1 = \{\{b_k^1\}_{k=1}^K, u^1\}$, $\mathcal{M}^2 = \{\{b_k^2\}_{k=1}^K, u^2\}$ are unimodal brief masses and b_k^1, b_k^2 is the belief mass assigned for k -th class according to each modality. u^1, u^2 and u are unimodal and joint uncertainty respectively. Thanks to the property of D-S theory, the final prediction of TMC will depend on the modality with low uncertainty. And thus the classification robustness is improved by integrating evidence from each modality. Inspired by TMC, researchers deployed SL-based uncertainty-aware multimodal fusion in many scenarios, including medical image classification [149], object detection [148], [158], semantic segmentation [2]. In whole-slide medical image classification task, by integrating evidences from modalities of multiple scales, they significantly improve the medical classification performance in autoimmune diseases detection and Fibroma classification tasks. Similarly, Chen *et al.* [159] also leverage Dirichlet distribution to model the uncertainty of multimodal representations and then fuse the segmentation and classification representations adaptively. Although TMC exhibits a great performance, Liu *et al.* [150] identified its limitations, including inability

关键思想是利用自注意力机制将不同模态的信息融合到一个共享的嵌入表示中，以在时间和空间上建模跨模态关系 [141]。王 等人发现现有的Transformer变体可能会在多模态任务上稀释单模态的注意力权重，导致整体性能次优。为了克服这个问题，他们提出动态识别无信息token，并来自其他模态的聚合特征替换它们，以实现有效的融合 [142]。与之前主要关注在训练时间内增强多模态融合可靠性相比，杨 等人 [157] 提出了利用注意力来在测试时间内完全处理损坏的模态。设计了一种置信度感知损失来增强可信预测并减轻噪声的潜在影响。

5.3 不确定性感知动态融合

与基于直觉假设的启发式动态多模态学习方法形成对比，这些假设在实际中可能并不总是成立，基于不确定性的多模态融合作为一种更通用和原则性的方法，最近涌现出来以实现可靠的融合，它通常建立在坚实的基础之上，例如概率分布或信息论。

主观逻辑是一种在分类任务中常用的不确定性估计方法，也是一个将Dirichlet分布的参数与信念分布联系起来的框架。Dirichlet分布可以被视为一个分类分布的共轭先验分布。SL为不确定性推理提供了一种强大的工具，并基于不完整或不确定信息进行决策。受SL的启发，最近提出了一种基于主观逻辑和证据理论的可信多模态分类（TMC） [147] 方法。它通过动态集成不同模态与Dempster-Shafer (D-S) 理论，为多模态分类提供了一种新范式。以两种模态场景为例，多模态信念质量按以下方式计算：

$$\mathcal{M} = \mathcal{M}^1 \oplus \mathcal{M}^2, \quad (9)$$

$\mathcal{M} = \{\{b_k\}_{k=1}^K, u\}$ 在哪里是多模态的简短质量， $\mathcal{M}^1 = \{\{b_k^1\}_{k=1}^K, u^1\}$, $\mathcal{M}^2 = \{\{b_k^2\}_{k=1}^K, u^2\}$ 是单模态的简短质量， b_k^1, b_k^2 是根据每种模态分配给第 k 类的信念质量。 u^1, u^2 和 u 分别是单模态和联合不确定性。由于D-S理论的特性，TMC的最终预测将取决于不确定性较低的模态。因此，通过整合来自每种模态的证据，分类鲁棒性得到了提高。受TMC的启发，研究人员在许多场景中部署了基于SL的不确定性感知多模态融合，包括医学图像分类 [149]，目标检测 [148], [158], 语义分割 [2]。在整张切片医学图像分类任务中，通过整合来自多尺度模态的证据，他们在自身免疫性疾病检测和纤维瘤分类任务中显著提高了医学分类性能。类似地，Chen 等人。 [159] 还利用Dirichlet分布对多模态表示的不确定性进行建模，然后自适应地融合分割和分类表示。尽管TMC表现出优异的性能，但Liu 等人。 [150] 指出了它的局限性，包括无法

to guarantee an overall decrease in uncertainty during the fusion of diverse views and neglecting of consistency across views. To address these issues, they proposed an improved opinion aggregation framework, ensuring consistency across views and achieving more reliable predictions.

Entropy is the most natural and straightforward way to capture modal-wise uncertainty. Tian *et al.* [144] proposed an entropy-based multimodal fusion method, named Uncertainty-aware Noisy-Or Multimodal Fusion (UNO). Specifically, UNO first introduces several uncertainty measurements, including predictive entropy, mutual information based on Monte Carlo Dropout, as well as deterministic entropy. It then proceeds to combine these uncertainty metrics for fusion by selecting the most conservative (highest uncertainty) measurement and weighting the outputs from each modalities based on the uncertainty. Finally, a simple but effective noisy-or fusion scheme is deployed to fusing decisions from multiple modalities. In multimodal semantic segmentation task, UNO can be used to improve the robustness in the presence of various unknown input degradation. Similarly, Zhang *et al.* [143] proposed to leverage the entropy of prediction as a proxy to gauge the importance of each modality, and then dynamically weight and fuse the unimodal predictions. By introducing such test-time dynamic fusion mechanism, the proposed method can be aware of uncertainty raised upon the varying quality of different modalities, and further enhance the effectiveness of multimodal fusion.

Gaussian distributions are another commonly used way to model the uncertainty. For example, a multivariate normal distribution can be used to model the distribution of multimodal feature. The variance of estimated distribution are capable of representing modality-wise uncertainty. Based on this point, Geng *et al.* [151] proposed a novel unsupervised multimodal learning method, called Dynamic Uncertainty-Aware Network (DUA-Net). Integrating multiple modalities of noisy data in unsupervised settings is a challenging task. Traditional multimodal methods either treat each modality as equally important, or adjust the weights of different modalities to fixed values, which fail to capture the dynamic uncertainty in multimodal data. By integrating the intrinsic information from multiple modalities under the guidance of uncertainty estimates from a generative perspective, DUA-Net obtains noise-free representations. With the help of uncertainty estimates, DUA-Net can weight each modality of a single sample according to its data quality, thus fully utilizing high-quality samples or modalities while mitigating the influence of noisy samples or modalities. The proposed method is shown to be effective in various unsupervised multimodal learning tasks. For audio-visual emotion recognition, Tellamekala *et al.* [152] proposed Calibrated and Ordinal Latent Distributions (COLD) to model uncertainty for each modality. In this work, the variance of high-dimensional latent distributions is learned first for each modality as a measure of uncertainty. In object detection task, previous work [146] proposed a probabilistic ensemble technique (ProbEnsemble) to combine the predictions from multiple detectors, each using different input modalities and architectures. In this work, the authors took the form of a Gaussian with a single variance to measure the uncertainty of unimodal predictions. Then, they probabilistically fused

保证在融合不同视图时不确定性整体下降，以及在视图之间忽略一致性。为了解决这些问题，他们提出了一种改进的意见聚合框架，确保视图之间的一致性，并实现更可靠的预测。

熵是捕捉模态间不确定性的最自然和直接的方式。

Tian 等。 [144]提出了一种基于熵的多模态融合方法，命名为不确定性感知噪声或多模态融合（UNO）。具体来说，UNO首先引入了多种不确定性度量，包括预测熵、基于蒙特卡洛Dropout的互信息以及确定性熵。然后，通过选择最保守（不确定性最高）的度量，并根据不确定性对每个模态的输出进行加权，将这些不确定性指标结合起来进行融合。最后，采用一种简单但有效的噪声或融合方案来融合来自多个模态的决策。在多模态语义分割任务中，UNO可以用于提高在各种未知输入退化情况下的鲁棒性。类似地，Zhang 等。 [143] 提出利用预测的熵作为衡量每个模态重要性的代理，然后动态地加权并融合单模态预测。通过引入这种测试时动态融合机制，该方法可以感知不同模态质量变化所引发的不确定性，并进一步增强多模态融合的有效性。

高斯分布是另一种常用的表示不确定性的方法。例如，多元正态分布可用于表示多模态特征的分布。估计分布的方差能够表示模态间的 Uncertainty。基于这一点，Geng 等 [151]提出了一种新的无监督多模态学习方法，称为动态不确定性感知网络（DUA-Net）。在无监督设置中整合多个模态的噪声数据是一项具有挑战性的任务。传统的多模态方法要么将每个模态视为同等重要，要么调整不同模态的权重为固定值，这无法捕捉多模态数据中的动态不确定性。通过在生成式视角的不确定性估计指导下，整合来自多个模态的内在信息，DUA-Net 获得无噪声的表示。借助不确定性估计，DUA-Net 可以根据单个样本的数据质量为其每个模态加权，从而充分利用高质量的样本或模态，同时减轻噪声样本或模态的影响。所提出的方法在各种无监督多模态学习任务中显示出有效性。在音视频情感识别任务中，Tellamekala 等。 [152]提出了校准和序数潜在分布（COLD）来为每个模态建模不确定性。在这项工作中，首先为每个模态学习高维潜在分布的方差作为不确定性的度量。在目标检测任务中，先前工作 [146] 提出了一种概率集成技术（ProbEnsemble）来组合来自多个检测器的预测，每个检测器使用不同的输入模态和架构。在这项工作中，作者采用具有单一方差的高斯形式来衡量单模态预测的不确定性。然后，他们概率性地融合

outputs from multiple modalities following Bayes' rule. The proposed probabilistic ensemble framework can estimate the likelihood of each detection proposal and then combine them into a more accurate and robust detection result.

Normal-inverse Gamma distribution hierarchically characterizes the uncertainties with parameters of high-order distributions. In the field of multimodal regression, a recent paper [153] presents a new multimodal fusion approach based on a Normal-Inverse Gaussian (NIG) mixture model. This NIG mixture model is similar to the decision-level fusion method, which combines multiple NIG distributions to output the uncertainty of each modality and the overall uncertainty. In this work, the NIG distribution is deployed to model modal-specific epistemic uncertainty (EU) and aleatoric uncertainty (AU). The contribution of each modality to the final prediction can be dynamically adjusted based on its quality and uncertainty, resulting in more reliable and accurate prediction. The proposed method is implemented using expert decision fusion and achieves significant improvements in both accuracy and uncertainty estimation in multimodal regression tasks.

Prediction confidence methods are designed to directly output the confidence. Different from the uncertainty estimation algorithms, confidence calibration methods aim to obtain confidence by directly calibrating the classification results without modeling the unknown data distributions. For example, the Maximum Class Probability (MCP) [160] can be regarded as a baseline method to obtain prediction confidence. Although effective in classification, MCP usually leads to over-confidence especially for erroneous prediction. Therefore, the True-Class Probability (TCP) [161] is employed to obtain more reliable classification confidence. Han *et al.* [154] proposed a multimodal classification method called Multimodal Dynamics that evaluates both feature-level and modality-level informativeness to selectively fuse multiple modalities in a trustworthy manner. Specifically, it uses a sparse gating strategy and true class probability estimation to assess the varying importance of features and modalities for each sample, achieving more reliable fusion when data quality varies across samples and modalities. Specifically, a modality-level gating strategy is employed to perform dynamic fusion as

$$\mathbf{h} = [\text{conf}^1 \cdot \mathbf{h}^1, \dots, \text{conf}^M \cdot \mathbf{h}^M], \quad (10)$$

where $[\cdot, \cdot]$ denotes concatenation operator, $\mathbf{h}^1, \dots, \mathbf{h}^M$ and $\mathbf{h}$ are unimodal and multimodal representations respectively. conf^m is the predicted confidence for the m -th modality which measures the informativeness. Zhang *et al.* [26] proposed a confidence-aware fusion method to selectively integrate features from RGB and thermal modalities for multi-spectral pedestrian detection. It predicts the classification confidence of each modality, then performs feature re-weighting to emphasize cues from more reliable modalities as well as suppress the less useful ones based on the confidence scores.

Theoretical justification has been proposed to shed light upon the advantage of uncertainty-aware dynamic multimodal learning. In multimodal learning theory literature, seminal work shows multimodal learning is provably better than unimodal from the perspective of generaliza-

多模态融合遵循贝叶斯法则的输出。所提出的概率集成框架可以估计每个检测提议的可能性，然后将它们组合成一个更准确和鲁棒的检测结果。

正态逆伽马分布层次化地刻画了具有高阶分布参数的不确定性。在多模态回归领域，近期的一篇论文 [153] 提出了一种基于正态逆高斯 (NIG) 混合模型的新型多模态融合方法。该 NIG 混合模型类似于决策级融合方法，它结合多个 NIG 分布来输出每个模态的不确定性以及整体不确定性。在这项工作中，NIG 分布被用于建模模态特定的认知不确定性 (EU) 和偶然不确定性 (AU)。每个模态对最终预测的贡献可以根据其质量和不确定性动态调整，从而得到更可靠和准确的预测。所提出的方法采用专家决策融合实现，并在多模态回归任务中显著提升了准确性和不确定性估计。

预测置信度方法旨在直接输出置信度。与不确定性估计算法不同，置信度校准方法旨在通过直接校准分类结果来获得置信度，而无需对未知数据分布进行建模。例如，最大类概率 (MCP) [160] 可以被视为一种获取预测置信度的基线方法。尽管在分类中有效，MCP 通常会导致过度置信，尤其是在错误预测的情况下。因此，采用真实类概率 (TCP) [161] 来获得更可靠的分类置信度。Han 等。[154] 提出了一种名为 Multimodal Dynamics 的多模态分类方法，该方法评估特征级和模态级的启发性，以选择性地融合多种模态，从而以可信的方式实现融合。具体而言，它使用稀疏门控策略和真实类概率估计来评估特征和模态对每个样本的 varying importance，当数据质量在样本和模态之间变化时，实现更可靠的融合。具体而言，采用模态级门控策略来执行动态融合，

$$\mathbf{h} = [\text{conf}^1 \cdot \mathbf{h}^1, \dots, \text{conf}^M \cdot \mathbf{h}^M], \quad (10)$$

where $[\cdot, \cdot]$ 表示连接运算符， $\mathbf{h}^1, \dots, \mathbf{h}^M$ 和 $\mathbf{h}$ 分别表示单模态和多模态表示。 conf^m 是第 m 个模态的预测置信度，用于衡量信息量。Zhang *et al.* [26] 提出了一种基于置信度的融合方法，用于选择性地整合 RGB 和热成像模态的特征，用于多光谱行人检测。该方法预测每个模态的分类置信度，然后执行特征重新加权，根据置信度分数强调来自更可靠模态的线索，同时抑制不太有用的线索。

理论依据已被提出，以阐明不确定性感知动态多模态学习的优势。在多模态学习理论文献中，开创性研究表明，从泛化角度来看，多模态学习比单模态学习具有可证明的优势。

tion ability [162]. Similarly, recent work [145] illustrates that uncertainty-aware dynamic multimodal learning can establish a tighter generalization error bound compared to traditional static methods. Combining with ensemble-like late fusion strategy in classification setting, we can perform uncertainty-aware weighting fusion in decision-level according to the following rule

$$f(x) = \sum_{m=1}^M w^m(x) \cdot f^m(x), \quad (11)$$

where $f^m(x)$ denotes unimodal prediction on modality m , w^m is uncertainty-aware fusion weights. Let $\hat{E}(f^m)$ be the unimodal empirical errors of f^m on D_{train} . Then with probability at least $1 - \delta$ ($1 > \delta > 0$), it holds that

$$\text{GError}(f) \leq \underbrace{\sum_{m=1}^M \mathbb{E}(w^m) \mathbb{E}(f^m)}_{\text{unimodal loss}} + \underbrace{\sum_{m=1}^M \text{Cov}(w^m, l^m)}_{\text{uncertainty-awareness}} + \epsilon, \quad (12)$$

where the first term on the right hand side denotes weighted unimodal loss, and the second term is the covariance between fusion weight and loss which represents the uncertainty-awareness ability of the fusion weights. Noted that given an effective uncertainty estimator, the covariance should be negative. This theorem demonstrates that uncertainty-aware dynamic fusion enjoys better generalization ability and further implies a principle to design novel dynamic multimodal fusion methods.

5.4 Discussion

Dynamic multimodal learning methods focus on the variance of modality quality with respect to sample, time or space, which widely exists but tends to be neglected. The dynamic fusion methods include heuristic (mostly designed for specific applications), attention-based (typically for representation fusion) and uncertainty-aware (modeling the modality and sample level uncertainty for fusion) strategies. There are great potentials for dynamic multimodal learning. First, the dynamic principle could be considered in SOTA multimodal models, e.g., CLIP. Second, there are lots of dynamic scenarios in real applications (e.g., autonomous driving, medical image fusion), so designing application-specific dynamic fusion strategies is interesting. For example, in multimodal medical images, we can dynamically fuse them at path-level, which may providing better flexibility and interpretability.

6 CONCLUSION

In this survey, we comprehensively investigate the main challenges of multimodal learning in the wild: noisy, partial, unbalanced, and quality-varying multimodal data. Some of them such as noisy multimodal data have been studied for a long time, but more recent interests in representation and translation have led to a large number of new multimodal algorithms and exciting multimodal applications. For example, low-quality multimodal data for foundation models, e.g., CLIP. (2) low-quality multimodal data for other tasks beyond fusion, e.g., alignment (retrieval). (3) Other low-quality forms

分类能力 [162]。类似地, 最近的工作 [145]表明, 不确定性感知的动态多模态学习与传统静态方法相比, 可以建立更紧密的泛化误差界。在分类设置中结合类似于集成式的后期融合策略, 我们可以根据以下规则在决策级执行不确定性感知的加权融合

$$f(x) = \sum_{m=1}^M w^m(x) \cdot f^m(x), \quad (11)$$

在 $f^m(x)$ 表示对模态 m , w^m 的不确定性感知融合权重。令 $E(f^m)$ 为 f^m 在 D_{train} 上的单模态经验误差。然后以至少 $1 - \delta$ ($1 > \delta > 0$) 的概率, 成立

$$\text{GError}(f) \leq \underbrace{\sum_{m=1}^M \mathbb{E}(w^m) \mathbb{E}(f^m)}_{\text{unimodal loss}} + \underbrace{\sum_{m=1}^M \text{Cov}(w^m, l^m)}_{\text{uncertainty-awareness}} + \epsilon, \quad (12)$$

其中, 右侧第一项表示加权单模态损失, 第二项是融合权重与损失之间的协方差, 它表示融合权重的鲁棒性。注意, 给定一个有效的鲁棒性估计器, 协方差应该是负的。这个定理表明, 鲁棒性动态融合具有更好的泛化能力, 并进一步暗示了一种设计新型动态多模态融合方法的原则。

5.4 讨论

动态多模态学习方法关注模态质量在样本、时间或空间上的变化, 这类问题广泛存在但往往被忽视。动态融合方法包括启发式 (主要为特定应用设计)、基于注意力的 (通常用于表示融合) 和不确定性感知 (对模态和样本级不确定性进行建模以实现融合) 策略。动态多模态学习具有巨大潜力。首先, 动态原理可以考虑在SOTA多模态模型中, 例如CLIP。其次, 实际应用中有大量动态场景 (例如自动驾驶、医学图像融合), 因此设计特定应用的动态融合策略很有意义。例如, 在多模态医学图像中, 我们可以在路径级别动态融合它们, 这可能提供更好的灵活性和可解释性。

6 结论

在本调查中, 我们全面研究了自然场景下多模态学习的主要挑战: 噪声、部分、不平衡和品质变化的多模态数据。其中一些, 如噪声多模态数据, 已被长期研究, 但近年来对表征和翻译的兴趣导致了大量新的多模态算法和令人兴奋的多模态应用。例如, 用于基础模型 (如CLIP) 的低品质多模态数据。(2) 用于融合之外其他任务的低品质多模态数据 (如对齐 (检索))。(3) 其他低品质形式

beyond noise, incomplete, unbalanced, and dynamic (e.g., adversarial). We believe that our survey will help catalog future research papers and better understand the remaining unresolved problems facing multimodal machine learning.

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除了噪声之外，还包括不完整、不平衡和动态（例如对抗性）。我们相信，本调查将有助于收录未来的研究论文，并更好地理解多模态机器学习面临的尚未解决的问题。

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